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Author

Chayma CHAABANI

Theme

A markerless solution to track the facial features of video games actors using a high definition camera



Host Institution : Digital Imagery Research and Development Centre (CDRIN)

Adress: 608, avenue Saint-Rdempteur, Matane (Qubec) Canada G4W 0E1

Tel :+1 418 562-1240, ext. 4010

Responsable : Mr. Pierre BEDARD

Supervisor : Mr. Pierre BEDARD

Pedagogic advisor: Mr. Majdi JRIBI

Abstract

In the video game industry, motion capture can detect the movements made by real actors, in order to animate the 3D characters.
The most effective facial motion capture systems require the existence of markers glued onto the face of the actor, and a system of infrared cameras to identify the positions of these markers, which is a source of discomfort for the actor.
An alternative technique is to use one or more HD cameras, without infrared and without markers on the actor's face. Therefore, we must develop a library to create virtual markers, based on the movements detected by the cameras.
In this graduation project, we propose a solution for facial features tracking, by developing a set of virtual markers on the video games actors' faces using one high definition camera and motion detection algorithms.

Key words : computer vision, motion capture, virtual reality, face detection, facial features tracking.

Résumé

Dans l'industrie des jeux vidéo, la capture de mouvement permet de détecter les mouvements réalisés par des acteurs réels, afin d'animer des personnages en 3D.
Les techniques de capture de mouvements faciaux les plus performants, utilisent des marqueurs physiques collés sur le visage de l'acteur et un système de caméras infrarouges pour repérer les positions de ces marqueurs, ce qui représente une source d'inconfort pour l'acteur.
L'avenir est donc aux systèmes de capture qui utilisent une ou plusieurs caméras hautes définitions, sans infrarouge, et sans marqueurs positionnés sur le visage de l'acteur. Ainsi, il faut développer une librairie permettant de créer des marqueurs virtuels en fonction des mouvements détectés par les caméras.
Dans ce projet, on propose une solution pour le suivi des traits faciaux, en développant des marqueurs virtuels sur les visages des acteurs de jeux vidéo à l'aide d'une caméra haute définition et d'algorithmes de détection de mouvement.

Mots clés : vision par ordinateur, détection de mouvement, réalité virtuelle, détection de visage, suivi des caractéristiques faciales.

"You are capable of more than you know.
Choose a goal that seems right for you and strive to be the best, however hard the path.
Aim high, behave honorably, prepare to be alone at times, and to endure failure.
Persist!
The world needs all you can give."

Edward Osborne Wilson

Dedications

The successful completion of this graduation project was made possible through the contribution of a number of people.
To say "thank you" to all of them is not even enough to express my gratitude.

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To my mother, Jamila, who have always loved me unconditionally and taught me to work hard for the things that I aspire to achieve.

To the memory of my grandparents, may ALLAH have mercy on their souls.

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"And my success is not but through Allah.
Upon him I have relied, and to Him I return."
The Holy Quran, Surat Hud [verse 88]

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Abbreviations

CDRIN	**C**entre de **D**veloppement et de **R**echerche en **I**magerie **N**umrique
Cégep	**C**ollège d'**e**nseignement **gé**néral et **p**rofessionnel
UQAR	**U**niversity of **Q**ubec **a**t **R**imouski
MoCap	**Mo**tion **Cap**ture
MPEG	**M**oving **P**icture **E**xperts **G**roup
PCA	**P**rinciple **C**omponent **A**nalysis
LDA	**L**inear **D**iscriminan **A**nalysis
SSM	**S**tatistcal **S**hape **M**odels
ASM	**A**ctive **S**hape **M**odels
AAM	**A**ctive **A**ppearance **M**odels
MUCT	**M**ilborrow/**U**niversity of **C**ape **T**own
UML	**U**nified **M**odeling **L**anguage
OpenCV	**Open** source **C**omputer **V**ision library
GUI	**G**raphical **U**ser **I**nterface

General Introduction

The human visual system has efficient techniques to easily produce, from the observed images, a description of the shape, the features and the position of the objects in the world around us. In the field of computer science, artificial vision is the discipline that attempts to automate the process of visual perception using different concepts from image processing, neurophysiology and psychology.

Since the early psychological researches related to artificial intelligence, it has been generally believed that modeling the cognitive parts of the intelligence was more difficult than representing its perceptual components [4]. However, enabling a machine to be aware of the visual world with all its rich complexity and constrains is actually far from being a trivial task.

Based on the idea that the evidence of human perception can provide clues for computer vision [49], the researchers focused on simulating the human perception by using the output information of a video stream, in order to analyze the images, in almost the same way that humans can interpret their environment through their sights.

Nowadays, computer vision and image processing systems are used in many fields ranging from extracting information from microscopy or X-ray images for medical diagnosis, to the security field where computer vision methods are used for recognition and video surveillance purposes.

Video games industry, also, showed a great interest in the artificial vision domain since it can provide a natural and powerful human-computer interaction in the virtual environment. This considering especially, the advances in 3D video games that mix real and virtual elements to achieve realistic characters, in order to ensure the player's immersion.

In this context, the face is probably the most important part of a 3D character. In fact, in the virtual world, as in reality, the attention of the observers is focused mainly on faces, considering the fact that facial expressions are the most important channel of nonverbal communication.

In the framework of the Digital Imagery Research and Development Centre project to improve its facial tracking system, and build realistic video games characters, we will conduct the researches in this challenging topic during our graduation project, in order to obtain the computer science engineer diploma from the National School of Computer Sciences.
The final result to be achieved at the end of this internship, is a set of virtual markers able to track the facial features of video games actors.
In the purpose of getting a realistic facial animation, the real face of the video games actor will be scanned and rendered by the CDRIN team, then, the actor's facial expressions will be reproduced on his synthetic 3D head model. To do so, the positions of our developed virtual markers will be used as a reference.
During this project, our research will focus on three major directions; motion capture, face detection and facial features tracking, in order to develop a robust markerless facial features tracker.

The rest of this graduation internship report is structured in two main parts:
- Theoretical study : we start this study in the second chapter, where we present an overview of the background information and the state of the art related to our fields of research. Then, the third chapter explains the facial features tracking method that we chose as a solution.
- Software development : the beginning of the application development stage can be found in the fourth chapter where we describe in details the requirements analysis. While the application design is analyzed in the fifth chapter through the semi-formal methodology "UML".

Throughout the final chapter, we expose the results achieved by the facial features tracker application and we evaluate some tracking performances. And, the first chapter describes the details of the project's general context and introduces the CDRIN, our host institution.
Last but not least, the conclusion summarizes the main aspects of our work and exposes possible future works.

Chapter 1

The general context

Introduction

The interest of this first chapter is to situate the project in its general framework, and to set the context and the goals for our work.
In this chapter, we will present the host research and development centre, its services and areas of research. Then, we will take care to present the problem statement as well as the purpose of this project.

1.1 Presentation of the host centre

In this section, we will introduce the Digital Imagery Research and Development Centre which proposed and accommodated this project.[1]

1.1.1 General presentation

The Digital Imagery Research and Development Centre, is a research center that is affiliated with the Cégep situated in the city of Matane, in the Lower St.Lawrence region of Qubec.
This new[2] research center is equipped with the largest motion capture studio dedicated to research in North America. It works with small, medium and large companies to

[1] Almost all the information of this section is excerpted from [N1].
[2] The CDRIN was established on october 2011.

carry out research and technology transfer in the field of digital imagery, with particular emphasis on motion capture.
The center also provides training to meet the skill development and advanced training needs of professionals in this rapidly evolving sector.



FIGURE 1.1: A scene of motion capture in the CDRIN. (Source: [N1])

1.1.2 Services

The Digital Imagery Research and Development Centre can offer the following services:

- Technical and technological assistance.
- Assistance in obtaining research funds.
- Access to technology testing facilities.
- Support for experiments conducted using the CDRIN facilities.
- Training programs related to research and development.

1.1.3 Areas of research

Three main areas of research are covered by the CDRIN:

Digital Imaging (Area developed with UQAR)

- Computer vision
- Development of hardware and software related to digital imaging

Motion capture and analysis

- Optimization of motion capture systems
- Hybridization of motion capture systems (hardware and software)
- Development of innovative processes to analyze and retrieve digital data
- Optimization of motion capture process efficiency

Human-computer interface, immersion and virtual reality

- Optimization and testing of the production steps involving interfaces
- Experimentation with hardware-software interfaces
- Application of interface technologies in health care
- Application of interface technologies in engineering and ergonomics
- Application of interface technologies in interactive media

1.1.4 Equipment

The Digital Imagery Research and Development Centre has the following equipment:

- Six motion capture systems:
  - 72-camera system with Vicon T160 cameras, the most accurate high-performance passive optical technology in its class.
  - 16-camera system with Vicon Bonita cameras(8 Bonita 3 and 8 Bonita 10), passive optical technology.
  - Animazoo IGS-190M system, inertial gyroscopic motion capture at its best.
  - Animazoo Gypsy 7 system, mechanical exoskeleton technology.
  - Vicon Motus system, video capture technology.
  - Organic Motion marker-less mocap system (using 18 video cameras)
- Ten workstations with high-performance PC or Apple computers.
- Editing workstation with Apple computer and Autodesk Digital Entertainment software.
- Autodesk 2013.

- The STAR PURSUIT SYSTEM 1000 from QUATTRIUUM that is a special edition system for the CDRIN, and combines motion capture and robotics, pushing back the limits of digital immersion and creating a new form of technology: motion tracking or MOTRACK. By combining robotics with localization, the capacities of both technologies are greatly improved. On the one hand, thanks to the mechanical agility of the mobile PTZ cameras, the MOCAP device is now free from the rigid grid of spatial coordinates required by most optical systems. On the other hand, multimedia robotic applications can now have greater spontaneity, free from choreographic restrictions, allowing direct interaction with mobile and dynamic environments. The QUATTRIUUM system also stands out in terms of technology as the first localization system whose primary image acquisition system is based on radio frequency technology. It is an adaptable system in which hybridization of acquisition technologies make it possible to obtain an optimal response in a multitude of applications.



FIGURE 1.2: The mocap of two video games actors in the CDRIN studio using the Vicon system. (Source: [N1])

1.2 Problem statement

The current motion capture system used in the CDRIN, is known as a passive mocap technology. It is based on several fixed cameras, that require spatial references to save the movements of the video games actor. To give this spatial information, actors have

to wear a special black suit containing physical markers.
However, these markers are harmful to the actor who wants to make unusual movements. Moreover, this system uses near infrared cameras to identify the positions of the markers, which is an additional source of discomfort for the actor. Finally, using this system to capture facial motions, is an unfavorable idea because of the complex movements produced by the human face. Especially, with the need to capture high degrees of emotion.
Consequently, the best alternative for facial motion capture, is to use markerless systems without infrared. In fact, this mocap technology uses one or more conventional high definition cameras, and requires the existence of a developed library to create markers on the actors faces. With these virtual markers, the CDRIN will be able to use the softwares that are already on the market, such as MotionBuilder, which is extremely powerful and widely used by the industry of video games and films. Actually, MotionBuilder is used to implement the saved motion capture data on the 3D character, in order to start the animation step.

1.3 Project presentation

The Digital Imagery Research and Development Centre has all the technical skills related to motion capture. However, most of the programs developed by its engineers are focused on the hardware level to make the interaction between different systems of motion capture.
The CDRIN is aware of the technical limitation of its MoCap systems, so the centre decided to add a layer of intelligence in its computers.
This graduation internship aims to develop the aspect of artificial intelligence in the facial motion capture solutions of the CDRIN. In fact, we will develop virtual markers on the face of video games actors, using motion detection algorithms and one high definition camera.

Conclusion

In this chapter, the context of the project was presented by introducing the host centre, and exposing the problem statement and the objective of our work.
Thus, in order to help achieving this goal, it is necessary to introduce, in the next chapter, some basic concepts to understand the current issue. We will also present the level of knowledge and development achieved in the facial motion capture studies.

Chapter 2

History and literature review

Introduction

In this chapter, we will give an overview of motion capture techniques. Then we will describe the most popular methods used for detecting and tracking the facial features.

2.1 Motion capture

Menache defines motion capture as "*the process of recording a live motion event and translating it into usable mathematical terms by tracking a number of key points in space over time and combining them to obtain a single 3D representation of the performance*"[27].
In this section, we will write a brief review of some essential motion capture details that are related to our motion tracking work.

2.1.1 The uncanny valley effect

Providing 3D characters with a complex emotional personality is a fundamental goal in the computer graphics and video games industry. Actually, the emotions must be realistically expressed at the visual level.
However, generating an animation of virtual characters without revealing their artificial nature is a difficult task. This difficulty is associated with the hypothesis of the "uncanny valley" exposed by the professor Masahiro Mori in 1970 [30].

This concept, first introduced for the field of robotics, describes the human emotional response while observing lifelike non-human entities.

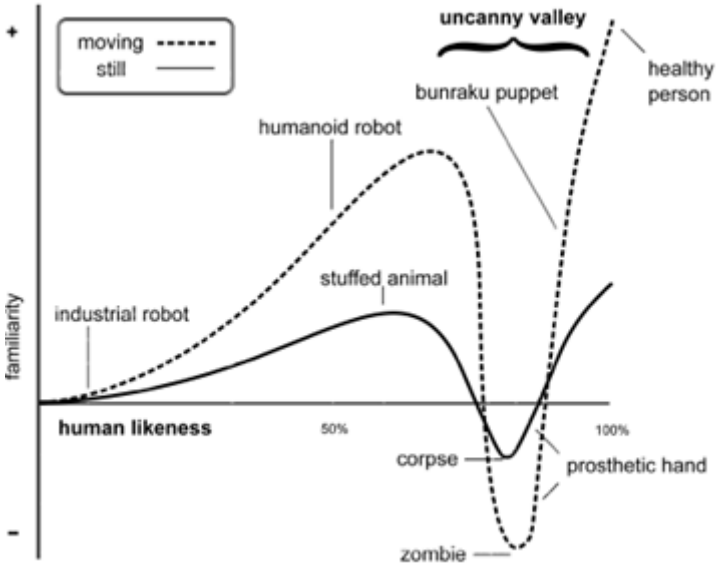


FIGURE 2.1: The "uncanny valley" plot. (Source : [30])

Figure 2.1 symbolizes as curves the emotional response of humans against the personification of an artificial character. The uncanny valley is the region of negative emotional response towards characters that seem "almost human". As shown in the figure, adding movement changes the shape of the uncanny valley by exaggerating the peaks and valley [41].

This theory has a direct impact on the philosophy behind the production of animated virtual characters. In fact, in fields like 3D modeling and motion capture, it is essential to use this concept wisely, in the purpose of ensuring that the characters are in their right places of the curve.

However, it stills very difficult to judge where to find a virtual character in this Mori's curve since it is difficult to know how each person will react, moreover, that requires testing the response of a large number of observers.

It is also difficult to avoid the uncanny valley when making a photorealistic human rendering. Over the course of the last decade, there have been numerous attempts, that have ended up somewhere in the depth of the uncanny valley.

More specifically, the facial animation is a particularly sensitive issue because of the complexity of its creation. It is well known that a major focus of research is to provide techniques to improve facial motion capture. This is a major goal which has not yet been achieved.

However, we mention in the following, titles of some 3D movies and video games that have partially succeeded in defeating the uncanny valley for facial animations:

–The digital Emily project (July/August 2010) :
The Digital Emily Project is a collaboration between the Image Metrics facial animation company and the Graphics Laboratory at the University of Southern Californias Institute for Creative Technologies (USC ICT) to achieve one of the worlds first photorealistic digital facial performances. The key technologies of this project included a fast high-resolution digital face-scanning process using USC ICTs Light Stage capture system and Image Metrics video-based facial-animation system [33].
The actress Emily O'Brien was placed at the center of a sphere containing 156 white LED and two stereo cameras (figure 2.2), in order to save 17 shots with different lighting directions to capture high-definition diffuse and specular components of the skin. This project also used a structured light to obtain a very specific geometrical data. From these data, a facial parameterization system based on blendshapes[1] was created, followed by the animation and the rendering process.
This project was able to render a synthetic facial performance which was generally accepted as being a real face. However, this quality is achieved with an amount of work and considerable efforts. Some timeframes and personnel required to achieve an animation of 90 seconds Full HD (1920x1080 pixels) were [32]:
- Scanning: 1.5 hours, 3 seconds per scan, 3 technicians.
- Scan Processing: 10 days, 37 processed scans, 1 artist.
- Rig Construction: 3 months, 75 blendshapes, 1 artist.
- Animation: 2 weeks, 90 seconds of animation, 2 animators.
- Rendering/Compositing: 3 months, 1 artist.
- Output: 24fps, 1920x1080 pixel resolution.



FIGURE 2.2: The actress Emily in the light stage during a scan. (Source: [33])

[1]Blendshape is a type of deformer that lets you change the shape of one object into the shapes of other objects.

–L.A. Noire (May 2011):

L.A. Noir, a Rockstar and Team Bondi investigative crime action game, was the first video game to use the MotionScan technology. It realistically maps an actor's face to the game engine, so very little post-processing work is needed. It enables a high resolution 3D recreation of a person's face. This technology uses 32 high resolution video cameras arranged as pairs to capture an actor's face during a live performance. The actor sits in a chair where the cameras are arrayed all around them [N8].

" *With performances that are captured with MotionScan, what you see is what you get, so there is literally no difference between what our actors have captured in the studio and what our players are interacting with in the game.*" [N9] says Brendan McNamara, founder of LA Noire developer Team Bondi.

As shown in figure 2.3, this work has achieved an important level of realism that was not feasible by conventional techniques.

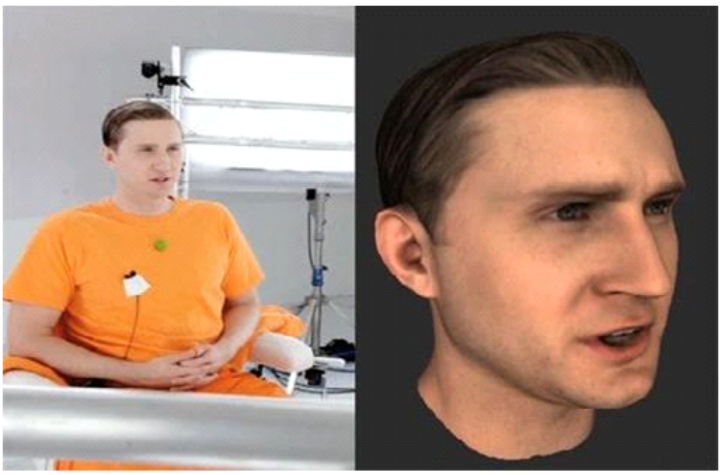


FIGURE 2.3: The motion scan result (Source: [N2]) : the real actor (left), the generated head model (right).

Based on the projects described above, we can notice that crossing the "uncanny valley" often requires a lot of time and expensive equipments.

2.1.2 MoCap systems

Two references in the field of motion capture are Eadweard Muybridge [31] and Etienne Jules Marey [5], who started their research by using the photography field to study the human and animal locomotion (figure 2.4).

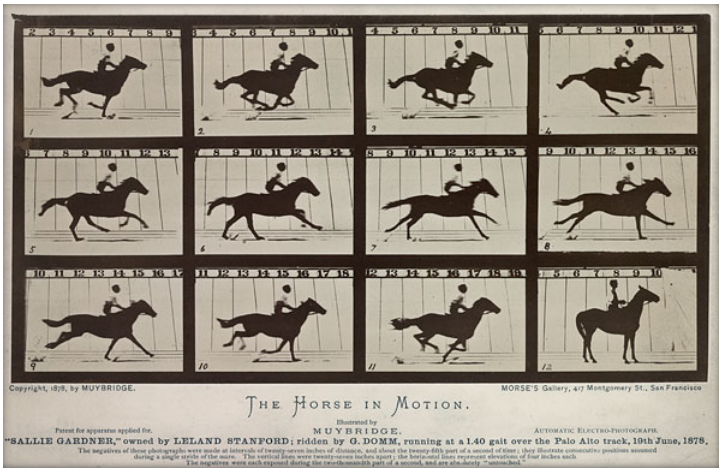


FIGURE 2.4: Horse in motion: excerpt from the animal locomotion study of Eadweard Muybridge.[1] (Source: [N3])

This experiment represented the first step of an intensive research work for motion capture systems.

The MoCap process can be divided into two parts:

- The body model estimation : this is the process that automatically determines a representation of the shape and the kinematic structure of the filmed person.

By shape we mean the geometric structure of the person, and the kinematic structure means both the accuracy of the joint and the length of the rigid parts (bone) structure.

- The acquisition of movement : this is the process that estimates the parameters of the mathematical representation, describing the movements made by a subject. This mathematical representation contains two components. The first describes the theoretical model of the kinematic structure of the body of the subject captured. The second represents parameters that are describing the movement relative to the representation chosen.

[1]Edward Muybridge first became involved in the photography of movement in 1872 when he was asked by Leland Stanford, the former Governor of California, to photograph his horse Occident. At the time, it was not known if a horse ever had all four feet off the ground while trotting. Muybridge was able to prove this true in his initial studies [51].

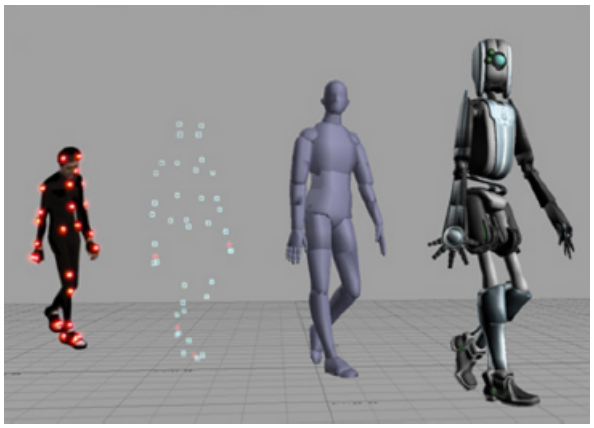


FIGURE 2.5: Motion capture principle, from the actor until the animatable model. (Source: [N12])

There is a variety of approaches which aim to solve these two problems. Some methods are mechanical, magnetic or inertial, but the most common are optical, using images or videos. Some approaches require the use of physical interaction structures such as exoskeletons, sensors for motion tracking, or optical markers. In the following, we will describe a number of the well-known methods used for motion capture.

2.1.2.1 Non-optical systems

During these last years, several types of MoCap systems were developed based on non-optical approaches and used some physical components.

- Inertial systems

Inertial technologies are based on the use of inertial sensors and sensor fusion algorithms. This approach offers an efficient Mocap method with a low acquisition cost.

Inertial movement data are wireless transmitted to a computer which computes and displays the movement performed in real time. Most inertial systems use inertial gyroscopes to measure rotations which are then transformed into joint movements.

This system does not require cameras, transmitters or external markers. It captures inertial motions and makes a real time estimation of the six degrees of freedom for each joint in the body.

The main interest of this approach is that the acquisition space is limited only by the wireless connection between the suit (figure 2.6) and the computer. This technology is particularly used in the industries of video games, for generating animations.



FIGURE 2.6: An example of the inertial mocap suit. (Source: [N4])

- Mechanical systems

Mechanical mocap systems directly follow the joint angles of the actor. This approach is often called exoskeleton motion capture technology.

An exoskeleton is generally a rigid structure made of metal or plastic, the joints contain potentiometers. These structures are often fragile and can interfere with the ground while doing complex movements.

A technician attaches the jointed structure to the body of the performer (figure 2.7). Therefore, when he moves, the sensors located at the joints measure his relative movements.

This type of wireless motion capture systems operates in real time and allow some limited acquisition space.

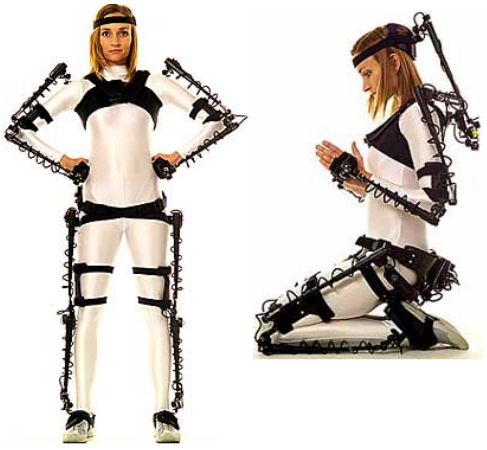


FIGURE 2.7: An example of the exoskeleton mocap suit. (Source[N5])

- Magnetic systems

Magnetic systems compute the position and orientation of the performer's body parts, by analyzing the relative magnetic flux of three orthogonal transmitter and receiver coils. The relative current of each coil allows to calculate both range and orientation of the movements.

This technology is not occluded by the actor's body parts. However, it is susceptible to the electromagnetic noise caused by the electrical interference from metal objects in the scene.

Moreover, the wires connected to the magnetic markers may control the freedom of the subject.

Finally the motion capture space, here, is generally limited, compared to the mechanical or inertial approaches.

2.1.2.2 Marker-based optical systems

In 1973, the psychologist Johansson was the first to do an experiment called "Moving Light Display" by attaching small reflective markers on human subjects to study the trajectory of joints[52].

Nowadays, this approach is the most commonly used by commercial MoCap systems. It uses physical illuminated points that are placed on the human to be tracked (figure 2.8). Pre-calibrated multiple high-speed cameras are used to shoot the movements of the performers.

The system tracks the position of the markers in the video stream and reconstructs their 3D triangulation trajectories. Due to self-occlusion, it happens that this mapping fails. It is therefore necessary to carry out a manual post-processing that smoothes the data and solves the problems of correspondence.

As popular as this technology has become, it still has many disadvantages. This existing system requires a large number of cameras and equipment, which makes it very expensive. Also, the markers required for tracking can be uncomfortable to wear and can have a significant effect on the actors freedom of movement. As a solution to these problems, methods of markerless motion tracking have started to be developed [18].

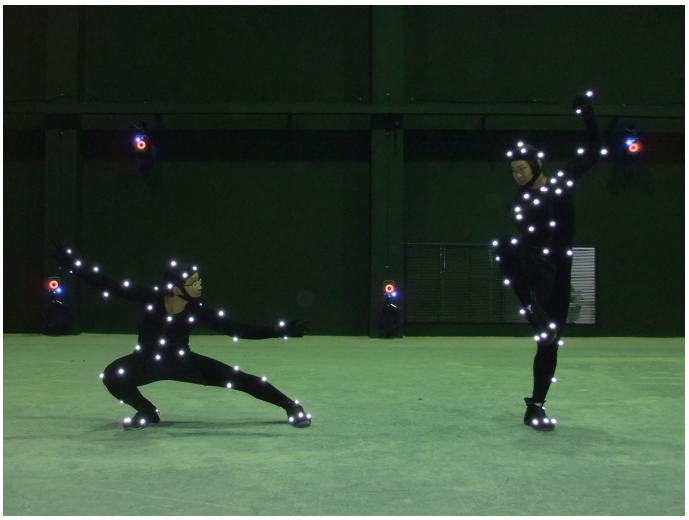


FIGURE 2.8: An optical mocap system: the little lights are placed on the body of the actors. (Source: [N6])

2.1.2.3 Markerless systems

Markerless motion capture refers to the task of full body motion capture without the need of markers or special suits attached to the body of the actor [6].
Over the past few years, several techniques has been proposed according to camera numbers and feature analysis. For example, in [18], they developed a markerless motion tracking system by using a skin detection method. Moreover, in [53], they used a gradientbased search to track a human subject with single and multiple cameras .

2.1.2.4 Facial motion capture

One of the first systems to track an actor's face and map it on to a computer generated model was introduced by Lance Williams in 1990. A 3D facial model was generated by taking a plaster relief of a subject's face, which was scanned and mapped to a cylindrical coordinate system. This research formed a benchmark for the future of facial performance capture [15].

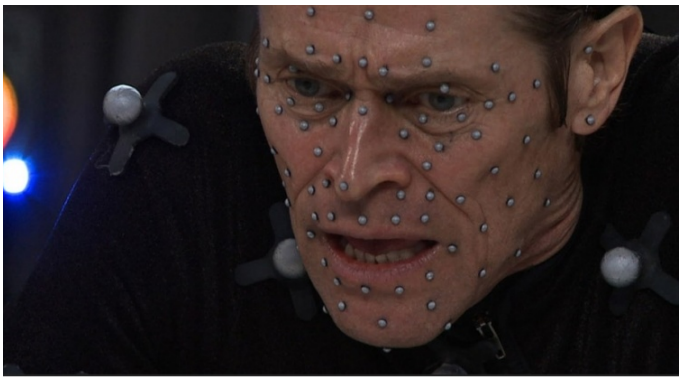


FIGURE 2.9: A marker-based system for facial motion capture. (Source: [N13])

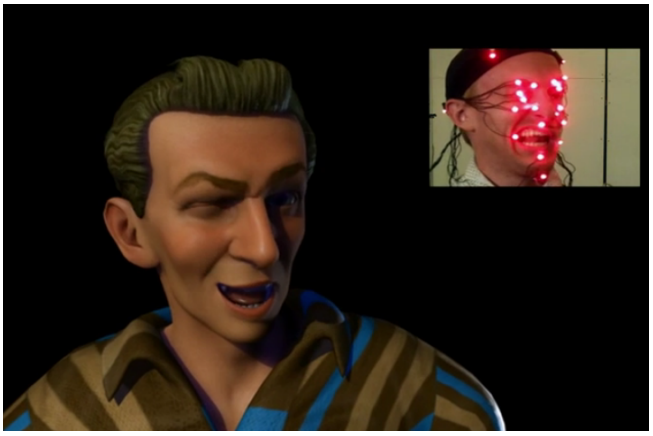


FIGURE 2.10: A marker-based mocap system for real time face tracking and animation: the face tracking(right), a 3D facial animation(left). (Source: [N7])

In this project, we will somehow follow this method, but we will create a marker-less motion capture system to track the facial features. Also, the 3D facial animation will be done at real time .
We will describe the details of this approach in the next chapter.

2.2 State of the art in analysis of Human faces

The first work on the automatic analysis of the face was in 1969 when Nagao and Fujibayashi proposed a system to detect the existence or absence of a face in a picture [29].
In the following sections, we will describe the face anotomy and a survey of face detection and facial features selection methods.

2.2.1 Face anatomy and coding systems

This part of the state of the art is devoted to the different systems of coding expressions. These systems are not routinely used by each of the existing methods of facial animation. However, it stills an important part to study for our project.

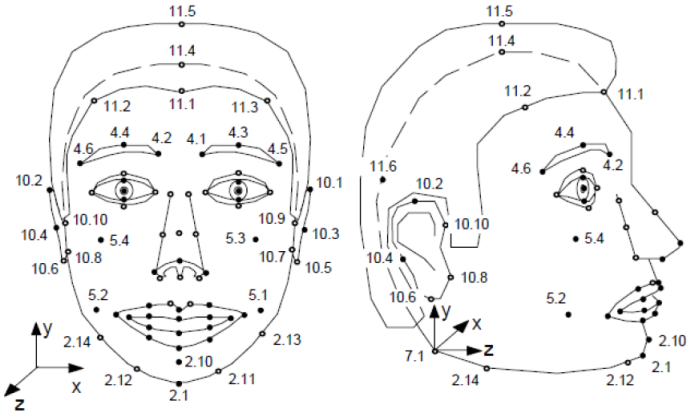
2.2.1.1 Facial Action Coding System (FACS)

The FACS decomposes the facial deformations into 46 action units (AU). Each action unit represents a change in the appearance and the facial muscles.
Facial expressions are then described as a combinations of action units and their intensities.
This coding system has been used in many studies related to the recognition of expressions. Among recent works, we can cite [14]. It has also been used in the generation of facial expressions and physical modeling of the face [48].
A comparative study of FACS coding approaches is available in [7].

2.2.1.2 MPEG-4 standard

In 1999, MPEG-4 defined a standard containing a section dedicated to the animation of the human body, including the face through a system of facial animation coding [34].
This coding system was specifically created to allow to different implementations, the use of a common protocol. The standard defines:
- 84 characteristic points (Facial Definition Parameters, FDP)
- 68 animation parameters (Facial Animation Parameters, FAP)
- 6 units based on anthropometric measurements of the face (Facial Animation Parameter Units, FAPU)

The FAP is the animation parameters that describe how the face is deformed, as well as the generation of head movements and expressions.
The FDP facial features points (figure 2.11) have two main roles: the first is to allow the measurement of facial features as well as the calibration of a generic model defined by data obtained from a real face and the second role is directly related the animation since FAP describes the displacements of FDP.
The FAPU are units of standardization based on distances and angles between certain characteristic points of the face for interpreting the values of FAP according to the morphology of the animated face.



Group	Number of FAPs
1: visemes and expressions	2
2: jaw, chin, inner lowerlip, cornerlips, midlip	16
3: eyeballs, pupils, eyelids	12
4: eyebrow	8
5: cheeks	4
6: tongue	5
7: head rotation	3
8: outer lip positions	10
9: nose	4
10: ears	4

FIGURE 2.11: The 84 FDP and FAP of the MPEG-4 standard. (Source : [34])

A lot of work was done based on this normalization of the facial animation. Some researchers have proposed parameterization methods for animating a face with the MPEG-4 [13]. Others used a model based on MPEG-4 to study the perception of emotions and expressions of a virtual character [26].
The facial coding systems mentioned above can be considered as an abstract parameterization of facial animation. The following section, will describe the existing methods of detecting and tracking the facial features.

2.2.2 Face detection

The first problem that needs to be addressed in facial motion capture is face detection. According to Ming-Hsuan Yang's survey [46], the challenges associated with face detection can be attributed to the following factors:
- Pose: the images of a face vary due to the relative camera-face pose, and some facial features such as eye or nose occlusion.
- Presence or absence of structural components: the possible presence of facial features such as beards, mustaches, and glasses may or may not be present and there is a great deal of variability among these components including shape, color, and size.
- Facial expressions because they affect directly the facial appearance.
- Occlusion: in an image with a group of people, faces may be partially occluded by other objects or faces.
- Image orientation: face images depend on the rotations of the camera's optical axis.
- Imaging conditions: when the image is formed, factors such as lighting and camera characteristics (sensor response, lenses) affect the appearance of a face.
Ming-Hsuan Yang [46] also classifies face detection methods into the following four categories:

2.2.2.1 Knowledge-based methods

The knowledge-based methods represent a top-down approach which uses a set of human-coded rules to guide the search process. These rules are constructed from the intuitive human knowledge of facial components. This approach works well for face localization in uncluttered background. However, to detect faces in complex backgrounds, Yang and Huang [45] used a hierarchical knowledge-based method which consists of three levels. The higher two levels are based on mosaic images at different resolutions, and the lower level uses an improved edge detection method.

2.2.2.2 Feature invariant approaches

This is a bottom-up method that detects structural invariant features. Facial features such as eyebrows, eyes, nose and mouth are commonly extracted using edge detectors. Then, based on the extracted features, a statistical model is built to describe their relationships and to verify the existence of a face.
These methods are based on the assumption that the structural features exist even when face orientation and lighting conditions change. We can mention one family of methods belonging to this approach: the methods based on skin color which has been used and proven to be an effective feature in many applications. Although people may have different skin color, studies [39] have shown that the major difference lies largely between their intensity rather than their chrominance.

2.2.2.3 Template matching methods

Template matching methods construct one or several templates in order to describe facial features. The correlation between an input image and the stored patterns enables the detection of the face. There are two types of correlation depending on the type of templates:
- Predefined face template: in this approach, eyes, nose, mouth and facial contours are defined by a segmentation method. This template matching method starts with a comparison between the images and the contours of the template to detect candidate faces. Then it translates the facial features to their candidate positions.
The detection here is performed in two stages: the first is the detection of candidate regions, the second is the examination details for determine the characteristics of the face. This method was first used in 1972 by [36].
- Deformable template: this approach is used in the purpose of modeling the facial features that adapt elastically over the model‘s face. In this method, facial features are

described using parameterized templates. A function is defined to connect the contours, vertices and edges in the input image, in order to match the parameters of the templates. The best adaptation of the elastic model is to find the energy function by minimising the parameters.
Template matching methods are simple to implement, however, they can‘t effectively deal with variation in scale, pose, and shape.

2.2.2.4 Appearance-based methods

In contrast to template matching, appearance-based methods use statistical analysis and machine learning techniques to discriminate between face and non-face in images.
In fact, the models are learned from a set of training images which capture the facial appearance and extract the relevant features of a face.
Some of the well-known machine-learning algorithms are Eigenfaces, Fisherfaces and Artificial Neural Networks (ANN). Both the Eigenfaces and Fisherfaces algorithms first calculate the average face that is the mathematical average of all the training images, so they can subtract the average image from each input facial image to have better face detection results. Yet, experimental results [3] demonstrate that the fisherface method has error rates that are lower than those of the eigenface technique for tests on the Harvard and Yale Face Databases. The most commonly used Artificial Neural Network machine-learning algorithm is the Multi-Layer Perceptron (MLP), which is a network of neurons with an input layer, output layer, and one or more hidden layers. Each layer has one or more neurons connected with the previous and the next layer.

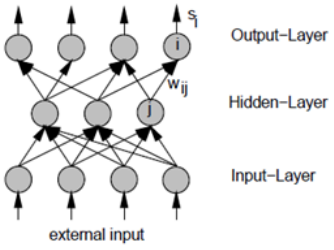


FIGURE 2.12: Topology of a typical feed-forward network with one hidden layer. (Source: [35])

Figure 2.12 shows the topology of a typical feed-forward network with one hidden layer. The external input is presented to the input layer, propagated forward through the hidden layer and yields an output activation vector in the output layer [35]. One of the earliest of methods using a supervised learning technique was the Viola&Jones method [42] which is able to detect the face effectively and in real time.

The purpose of this method is to create a classifier containing a set of facial characteristics. This classifier is constructed using haar-like features. The learning process through a large number of images. Viola and Jones choosed to use a classification by boosting which means using multiple learners called "weak" in order to create a "strong" classifier.
The Viola&Jones face detection algorithm was also the first to introduce the concept of integral image to computer vision. The value at any point (x,y) in the integral image is the sum of all the image pixels above and to the left of (x,y) inclusive. This new image representation allowed a rapid comptution of the haar-like features.



FIGURE 2.13: The detection accuracy of the Viola&Jones face detection algorithm. (Source:[23])

Figure 2.13 shows some images describing the detection accuracy of the Viola&Jones face detection algorithm. As we can see, this face detection algorithm achieves respectable detection rate.

2.2.3 Facial features extraction

A human can recognize facial features from his early childhood. This is an automated process in our brains. However, that represents a challenge to the computers. In fact, this feature extraction process can be defined as the extraction of relevant information from a face image. This information must be valuable to the later step of tracking.
There is much literature about extracting facial features methods, some of them will be discussed in the following sections. Nevertheless, we represent in Table 2.1 a list of facial feature extraction methods.

Table 2.1: Facial feature extraction methods (Source: excerpt from [22])

Approach	**Face detection**	**Feature points to extract**
Using active shape models	-	Flexible in term of number of features to track
Identification of the face symmetry axis and then feature points using correlation	Included	Location of eyes and mouth
Using adaptive Hough transform, template matching and active contour models	Included	Eyes, eyebrows, mouth, nostrils, cheeks and chin
Eye corner detection and template matching	-	Eyes
Using energy function to extract facial features	Included	Face region, iris, eyebrows, nose, mouth, eye corners, face-hair
Using active contour model and color information	Included	Face region, eyes, nose, mouth
Using Support Vector Machines and Multi wavelet decomposition	Included	Face region, eyes, nose, mouth
Using linear combination model	-	Eyes, eyebrows, nose, mouth
Using PCA and Wavelet multi resolution images	-	Eyes, eyebrows, nose, mouth

2.2.3.1 Principle Component Analysis

This is the reference algorithm for face analysis. PCA is an orthogonal linear transformation that transforms the data to a new coordinate system.
This method is also known as the Principal Component Analysis space reduction. In 1991, Turk and Pentland [40] used this transformation, when they introduced the "Eigenfaces" to express the different eigenvectors database of facial images of PCA.
Actually, we can consider that face images of NxN pixels are a subset of the image space of dimension NxN. Thus, it is important to find the representation of a face in this "face space". This new space is composed of a training database which is a set of face images. PCA aims to maximize the variance in this dataset by maximizing the variance between "different faces". Then, using data projection, the greatest variance comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.
The PCA approach is quite dependent on the chosen face space. Indeed, the more the training set is large and includes variations of illumination, pose and expressions, the better we can represent faces in this space.

2.2.3.2 Linear Discriminant Analysis

Linear Discriminant Analysis, first introduced by Belhumeur, Hespanha and Kriegman in 1997, helps to better understand the distribution of the feature data. It maximizes the ratio of between-class variance (differences of human features for the measurements of different persons) to the within-class variance (differences for the same person) in any particular data set thereby guaranteeing maximal separability [2].
This method easily handles the case where the within-class frequencies are unequal and their performances has been examined on randomly generated test data. However, the LDA approach can be seriously degraded if there are only a limited number of observations N compared to the dimension of the feature space n. To prevent this from happen is it is recommended that the linear discriminated analysis be preceded by a principle component analysis [20].
In [2], the author clarifies the difference between PCA and LDA methods as follows: in PCA, the shape and location of the original data sets changes when transformed to a different space whereas LDA doesn‘t change the location but only tries to provide more class separability and draw a decision region between the given classes.

2.2.3.3 Elastic graph matching

Local approach based on the comparison of local face features, is the most commonly used for real-time projects. The "Elastic Graph Matching" (EGM) algorithm that is based on the "Dynamic Link Architecture" (DLA) [21], uses this approach.
The EGM represents the face by a labeled graph (figure 2.14), which is a set of nodes connected by edges. Each node can correspond to a feature point of the face or to a node of a grid superimposed on the face. This representation enables us to have a local information across the nodes and an overall information through the inter-connections. The node is characterized by its name or its position and a vector that contains the analysis of its local neighbourhood. This information can be determined using an analysis by the Gabor filter [12], and the result of this analysis is stored in a vector called "Jet". The jets and the relative positions of the nodes form the model of the face.

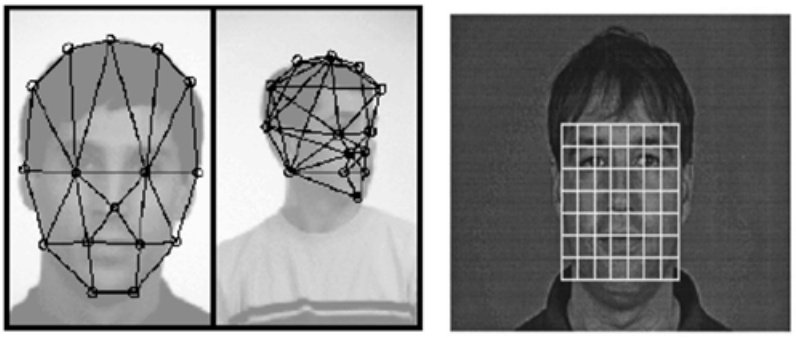


FIGURE 2.14: Reference grids for elastic graph matching. (Source: right [12], left [44])

2.2.3.4 Statistcal shape models

A Statistical Shape Model (SSM) is a statistical representation of a shape obtained from a training data. These models are constructed from a representative set of example shapes and represent the normal shape variations of a class of objects.
The most popular SSMs are: active shape models, and active appearance models.
• **Active Shape Models**
According to Kendall's definition [19]: shape is "*all the geometrical information that remains when location, scale and rotational effects are filtered out from an object.*"
Active Shape Model (ASM), first introduced by [11], is a statistsical approach for feature extraction and modeling the variability of the shapes for a given object.
This parameterized statistical shape representation requires a set of annotated training images in such a way that each landmark represents a distinguishable point present on

every image [8].



FIGURE 2.15: A manually annotated face image. (Source: [8])

The initial feature points are obtained from the mean shape, which is derived from the training data, and its accuracy depends on the size of the training set. In addition, the local structure of the feature points is represented by the change in intensity values of the pixels along a profile line (i.e. edge location) that goes through the feature points. This is based on the assumption that usually the facial features are located on the strong edges.

An example of ASM's performance was shown in the [38]'s thesis, where they used a training set of 22 silhouettes of faces that were taken by a digital camera then segmented and marked by hands. The generated shape model was able to capture 98% of the variation of the training set.

• **Active appearance models**

An active appearance model (AAM) is an algorithm for matching a statistical model of a deformable object to a new image. In fact, to account for texture variation, the ASM model has been expanded to statistical appearance models including an Active Appearance Model [10].

The proposed AAM combined a model of shape variation (ASM) with a model of the appearance variation of shape-normalized (shape-free) textures. The key idea of AAMs is to statistically model the texture and the shape of a deformable object.

According to [24], the basic AAM principles are:

- Modeling: AAMs are built during a training phase. The training phase requires a set of images with labeled landmarks. The shape, represented by a vector formed by concatenating the positions of labeled landmarks, is first normalized using Procrustes analysis and then projected on a low dimensional subspace using PCA.

Then, images in the training set are warped to the mean shape to produce shape-free

patches. The texture from these patches is normalized and placed in a vector. Again, the PCA is used to reduce the dimensionality.
- Model fitting: Once the model is created, it is interesting to optimize the parameters and fit the model to an object (in our case the object is the face) appearing in the new testing image. The task of AAM fitting is an optimization problem, which is often difficult to solve. It is usually addressed using a gradient descent algorithm or linear regression.

When the AAM method is applied to tracking, successful results can be obtained even when perspective, rotational and translational transformations are present. However, problems arise in the presence of nonlinear variations of texture and shape, fast motion and other special circumstances. Many improvements to the basic AAM algorithm have been proposed over the years to compensate for these deficiencies, such as the work done in [25] and [47].

2.2.3.5 Comparison between the facial feature extraction methods: advantages and disadvantages

In this section, we address the problem of facial features extraction. This choice has been done since it is necessary to extract the features before starting the tracking step. We discuss the pros and cons of the most popular facial features extraction approaches found in the literature. Table 2.2 shows the advantages and disadvantages of the methods that we've introduced above.

Table 2.2: Facial feature extraction methods: advantages/disadvantages

Methods	**Advantages**	**Disadvantages**
PCA	Correlation between components of the data vector be clearly viewed and it capture the second order correlation value. It uses the eigenface	Identifies the linear combinations of variables and Ignore the high order correlation value

Table 2.2 – *Continued from previous page*

Methods	**Advantages**	**Disadvantages**
LDA	Linear mapping, dimensionality of the subspace is limited by the number of classes of the data	Cannot handle data in which the individual classes are far from Gaussian, suffers from SmallSample Size problem
Elastic graph matching	Using the features only at the key nodes of the image reduces the noise taken from the background of the face images. Also, the graph matching problem can be optimized efficiently by linear programming.	Its complexity makes it very difficult to take into account all types of dependencies between the vertices and edges in the model and data graphs.
Active shape models	Performs exceptionally well when segmenting objects that do not have a clear or continuous boundary.	It takes a signicant amount of work to manually delineate training images and build the model
Active appearance models	A priori knowledge is done through observation of both shape and texture variation in a training set. Then, a description of the face is derived, which can be used to rapidly search images for new object instances.	Problems with size of multidimensional models

Conclusion

This chapter served at first as an explanatory of different motion capture methods. This study is essential in order to understand the general context of the topic. Furthermore, we introduced a state of the art related to face detection and facial features extraction. In the next chapter we will explain the method that we will use as a markerless facial features tracking solution.

Chapter 3

The facial features tracking solution

Introduction

This chapter describes the facial features tracking method that we will use as a solution. This method is similar in spirit to the statistical shape models approach. In fact, this tracking method consists of two components: data and algorithms. The algorithms typically operates on the incoming data (here video stream) by referencing stored data (training dataset) as a guide.

3.1 Annotate the dataset

The first step in building a facial features tracker is to create an annotation tool, in order to specify the locations of the desired facial features in each image of the training database.
The data here, is a collection of images that contain an entire face. For best results, these images should be specialized to the identity of the performer and the lighting conditions in which the facial features tracker is later deployed. It is also important that the faces in the training database exhibit the range of head poses and facial expressions that the intended application expects.
The larger the set of images is, the more robust the algorithms behave, as they become more aware of the variability that faces can exhibit.

In this research project, we developed an annotation tool that captures images from a HD video action camera, then invites the user to mark points and connections to specify the facial features that require tracking. However, there are some publicly available pre-annotated face databases that can be used for research or even commercial purposes.
Table 3.1 illustrates a comparison of some landmarked 2D face databases. Based on the statistics shown in Table 3.1, we notice that the MUCT database that consists of 3755 faces with 76 manual landmarks (figure 3.1), provides more diversity of lighting, head poses, age, and ethnicity than currently available databases. Therefore, the MUCT data should be suitable for training and generating a wide collection of models [28].
To ease the process of manually annotating a large collection of images, each with a large number of points, we will use the MUCT dataset as a generic (person independent) training database. Additionally, the annotation tool that we developed can also be used to optimize the performance for a given lighting and head poses settings.

TABLE 3.1: STATISTICS OF PABLICLY AVAILABLE LANDMARKED FACE DATABASES (SOURCE: EXCERPT FROM [28])

	MUCT	**AR**	**BioId**	**PUT**
Nbr of landmarked images	3755	508	1521	2193
Nbr of landmarks	76	22	20	199
Nbr of subjects	276	126	27	200
Image size (pixels)	480x640	768x576	384x286	2048x1536
Wearing glasses percent	18	31	32	0
Viola&Jones fail percent[42]	1.8	1.2	4.3	0.2



FIGURE 3.1: The 76 MUCT landmarks. (Source:[28])

3.2 Global shape transformations: Generalized Procrustes Analysis

Based on the shape definition mentioned earlier in chapter 2. section 2.2.3.3, we became aware that the building of a deformation model of face shapes must first be preceded by processing the annotated data to variations attributable to global transformations that include the scale (i.e. the size of the face in the images of the annotated database), in-plane rotation and translation of the face in the dataset.

To do so, it is desirable to align the shapes into a common coordinate frame using the generalized Procrustes analysis [1] [16], which is one of the statistical methods for comparing the shapes of objects.

The objective of this analysis is to adjust each data instance (i.e. face image) to fit a mean shape computed from that training database. This alignment is measured as the least square distance between the target shape and the mean shape.

[1]Procrustes was the nickname of a robber in Greek mythology called Damastes, who lived by the road from Eleusis to Athens. He offered travelers hospitality on a magical bed that would fit any guest. His humor was to stretch the ones who were too short to fit the bed until they died or, if they were too tall, to cut off as much of their limbs as would make them short enough. This rather unpleasant practice continued until Damastes was killed by Theseus, son of thra and the Athenian king geus. Another nickname for Damastes was "The one who stretches". The term Procrustes Analysis was coined by Hurley&Cattell in 1962 [37].

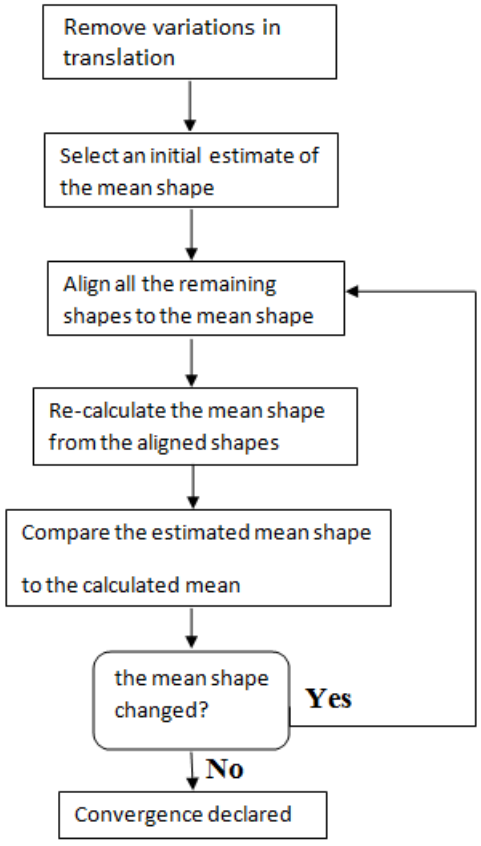


FIGURE 3.2: Schema describing the algorithm of procrustes analysis

The mean shape, mentioned in the algorithm described in figure 3.2, indicates the mean of all landmark points in an entire set of annotated images. The first shape in the database is chosen to be the initial estimate of the mean shape. Whereas, the mean shape calculated in each iteration, is computed by normalizing the average of all shapes so that $\| x\| = 1$.
The normalization here is necessary to fix the size and the orientation of the mean shape at each iteration.

After this Procrustes alignment, the directions of variability stemming from rigid transformations are removed from the data. Also, the feature clustering becomes more compact and their distribution becomes more representative of the facial deformation.

3.3 Compact shape representation : PCA

After removing the rigid transformation of the facial geometry, we were left with a set of landmark points aligned to the mean shape. However, it remains the local deformation that represents the variability of facial shapes across identities and expressions.

In this section, we will represent these facial deformations in a compact parametric way, so that the most important features can be extracted easily.
The simplest method that has a low computational complexity, is the linear representation. Assuming linearity, the problem now is to find the appropriate change of basis. The main idea here, is to find a low-dimensional subspace embedded within the space that contains all the valid facial shapes.
It is important to know that the lower the dimensionality of this subspace, the more compact the representation of the face shape is, and this leads to more robust features tracking. At this step, we are using the PCA method to find the best low-dimensional subspace. The input data here is not the annotated face images, but the matrix "S" of the shapes described by landmark points.
PCA will find an orthonormal matrix P that projects S in this new subspace. The rows of the P matrix are the principal components of the covariance matrix of S.
Let S be a M x N matrix, where m is the number of facial feature types and n is the number of the mean Procrustes-aligned shapes. Then, the cov(S), which is a M x M matrix, can be computed as follows:

$$\text{cov (S)} : \text{C} = \text{S}^t S$$

The eigenvectors and eigenvalues of the matrix C are calculated using the Singular Value Decomposition (SVD) method. For more linear algebra information about this operation, please see Appendix A.
The PCA result
The variance in the major directions is ordered from largest to smallest, so that the eigenvector corresponding to the largest eigenvalue represents the direction of the first principal component, the second eigenvector corresponds to the second principal component, and so on.
A common approach to choose the dimensionality of the subspace is to select the subspace by evaluating the variance in the top k[1] directions of variability (eigenvectors).
After applying the PCA, we obtain the following statistical shape model representation:

$$S = \overline{S} + P_s b_s \tag{3.1}$$

The vector $\overline{S}$ contains the mean shape, Ps is a matrix of the first eigenvectors (that represent the directions of variation derived from the training set) of the covariance matrix and bs is a vector that defines the shape parameters. The variance for each mode of variation is given by the eigenvalue λ_i.

[1]The choice of the "k" factor depends on the size of the data and the available computational power.

By varying the parameter bs, we can vary the shape, however, based on the [22] results, we can limit the parameter in the range of $\pm 3\sqrt{\lambda_i}$ to ensure that the generated shape is similar to those in the original training set.

3.4 Modeling the texture : Building the gray-level model

In order to fully describe the variation of the face, both its shape and texture must be modeled. In this section, we will almost follow the shape model building method.
The shapes were aligned, but they were not identical and there was no guarantee that pixels with same coordinates represent the same part of the face from image to image in the training set. Therefore, the first step consists on warping each training image to the mean shape, to obtain a shape-free patch. That means modifying an image so that its landmark points match the mean shape in order to remove any variation in appearance due to shape differences. This can be done using the method described in appendix B. In fact, the idea is very simple: we create triangles including our annotated points and then iterate through these triangles in order to warp one triangle from an original annotated image into a generated distorted one.
After wrapping all the training images to the mean shape, we obtain texture vectors g with the same length as the mean-shape vector Sm. Then, we focus on reducing their variation in lighting conditions by normalizing these shape-free images. The idea here is similar to the procrustes analysis method (described in this chapter, section 3.2) that removes the global shape variations from the training set of images.
This normalization process is done using Algorithm 1.
After that each texture vector value is normalized to one, and its mean value is subtracted, the vectors are then placed as columns in a texture matrix T. As for the modeling of shape variations, the next step is to describe the variability of texture of human faces. To do so, Principal Component Analysis is applied again, but with the use of the eigenfaces that contain the texture variation of the images.
The final model is represented as follows :

$$g = \overline{g} + P_g b_g \tag{3.2}$$

Where $\overline{g}$ is the mean gray-level vector, Pg is a set of orthogonal modes of texture variation, and b contains the eigenvalues that represent the texture parameters.

[1] I is a vector of 1 and has the same size as the texture vector g.

Algorithm 1 Algorithm to normalize the texture vectors

INPUT: Set of texture vectors $g_i = (x_1, x_2, ...x_n)_i$ wrapped to the mean shape.
OUTPUT: Normalized texture vectors.
INSTRUCTIONS:

1. Choose the first vector in the set of texture vectors as an initial estimation of the mean, $\overline{g} = g_0$.

2. Normalize all the texture vectors g_i by applying the brightness α and the contrast β factors that best match the vector of the estimated mean texture.

$$g_{inormalized} = \frac{g_i - \beta * I^1}{\alpha} \text{ and } \alpha = g_i.\overline{g}, \beta = \frac{\sum_{j=1}^{n} x_j}{n}$$

3. Calculate the mean of the new set of normalized texture vectors
4. Normalize the new mean using the α and β factors, so that the resulting mean vector has zero mean and unity variance.
5. Compare the new mean to the estimated mean texture.

if the mean changes **then**
    return to step 2
else
    algorithm converged
end if

3.5 Modeling the shape-texture correlation: Building the Combined model

In the previous sections, we built two separate statistical models of facial shape (expression (3.1)) and texture (expression (3.2)). It is then interesting to combine them into one model that inherits all their benefits and uses much less parameters.
Since that bs and bg are the shape and the gray-level (texture) parameters, for each training image, we will compute the following concatenated vector.

$$\mathrm{b} = \begin{pmatrix} W_s b_s \\ b_g \end{pmatrix} = \begin{pmatrix} W_s P_s^t (s - \overline{s}) \\ P_g^t (g - \overline{g}) \end{pmatrix}$$

$\mathrm{W}_s$is a diagonal matrix of weights for each shape parameter. In fact, these weights are essential here to take into account the difference in units, since the bs is measured in pixel distance and the b_g has the pixel intensity unit. A simple method to estimate W_s is to use the ratio of the total variance in shape and texture.

$$\mathrm{W}_s = rI = \begin{pmatrix} r & 0 & ... & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & ... & 0 & r \end{pmatrix} \textit{ and } r = \frac{\sum \lambda_s}{\sum \lambda_g}$$

In the next step, PCA is applied on the vector b and we get:

$$b = Qc, Q = \begin{pmatrix} Q_s \\ Q_g \end{pmatrix} \tag{3.3}$$

Here, the eigenvectors matrix Q combines $Q_s$and Q_g which represent respectively the eigenvectors describing the shape and the texture variability, and c is the set of parameters. Finally, the shape and texture variation can then be represented as follows:

$$\mathrm{s} = \overline{s} + P_s W_s Q_s c$$
$$\mathrm{g} = \overline{g} + P_g W_g Q_g c$$

3.6 Model initialization : Face detection

At this stage, we still have the problem of model initialization in the first frame of the sequence. An obvious solution for this is to use a face detection method to append the model on the face position. The Viola&Jones algorithm introduced in the last chapter (section 2.2.2.4) can do this efficiently[23].
The Viola&Jones method runs a detection sub-window of 24 x 24 pixel over the images, in order to detect the presence of a face. Once the sub-window has fully covered the image, the sub-window size is increased using a scale factor of 1.25 and it restarts the scanning of the image again. The algorithm comes to an end when the detection sub-window will have the same size of the image. Scanning the image is simply done by shifting the sub-window a pixel to ensure the maximum accuracy.
A preliminary and very important step of this method is to train the classifier. This is about training the classifier to make it aware that we want to detect faces. To do so, the classifiers undergo two types of situations the first where a huge number of positive examples (face images) are presented and the second with a huge number of negative cases (non-face images). In their original article [42],Viola and Jones have trained their classifier using an image database of MIT. The result was a .xml file classifier.
It is also important to know that the features used here are the haar-like features (figure 3.3) which are a synthetic and informative representation calculated from the pixel values using the integral image method. The use of the integral image representation saves the time calculation of features, since it contains in each of its points the sum of the pixels located around the current pixel.

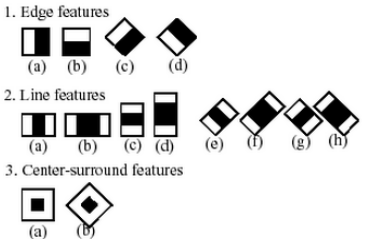


FIGURE 3.3: The haar-like features. (Source: [42])

Now comes the last part of the algorithm: "boosting for feature selection" (figure 3.4) that uses several "weak" combined classifiers rather than using a single "strong" classifier. In this way, Once a classifier decides that there is no face, the sub-window is immediately rejected which saves a considerable treatment time.

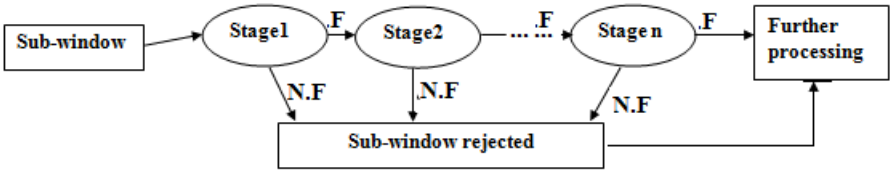


FIGURE 3.4: The boosting selection process : F indicates the detection of a face and N.F refers to the detection of a non-face.

To summarize, figure 3.5 describes this face detection algorithm:

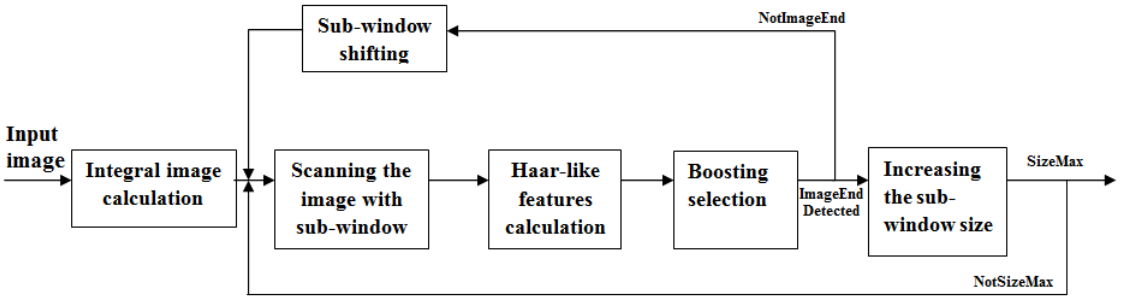


FIGURE 3.5: Schema describing the Viola&Jones algorithm.

3.7 The tracking process

In the previous sections, we built a combined model of shape and texture variations from the training images (expression (3.3)). Then, we explained the face detection method. In the tracking process, we will focus on matching the constructed model with the input

face images illustrated in the video stream. The objective here is to find the parameters that best fit the model to the input unseen image (i.e image that was not in the training dataset).
This fitting can be considered as an optimization problem because we are seeking for the parameters that minimize the differences between the model and the new input image, and bring the instantiated model as close as possible to the input image.
This problem can be represented by the following expression :

$$f(x) = I_{input} - I(x)_{generatedModel}$$

$x = (x_1, x_2, ..., x_n)$ denotes the unknown vector of the parameters that minimize the function.
An efficient approach to find the global minimum of this function, is the stochastic (non-deterministic) gradient descent algorithm which is a minimization method for multidimensional functions. The search strategy of this method can be divided into two ideas:

- Save all the local minima of the function.

- The global minimum is the smallest value of the minima saved in the previous step.

Finding the local minima can be done by running through the graph of the function f(x) as described in the following steps[1]:

1- Choose an initial starting point x_0.

2- Select the next point $x_{k+1} = x_k + d_k g_k$, *and* $g_k = -\nabla f(x_k)$.
Here, g_k is the opposite of the gradient of the function and it denotes the descent direction, and d_k is the fixed step size.

3- if ($f(x_{k+1}) < f(x_k)$)

save x_{k+1}

Else return to step 2.

[1]More mathematical details about this method can be found in[50].

3.8 Outline of the facial features tracking solution

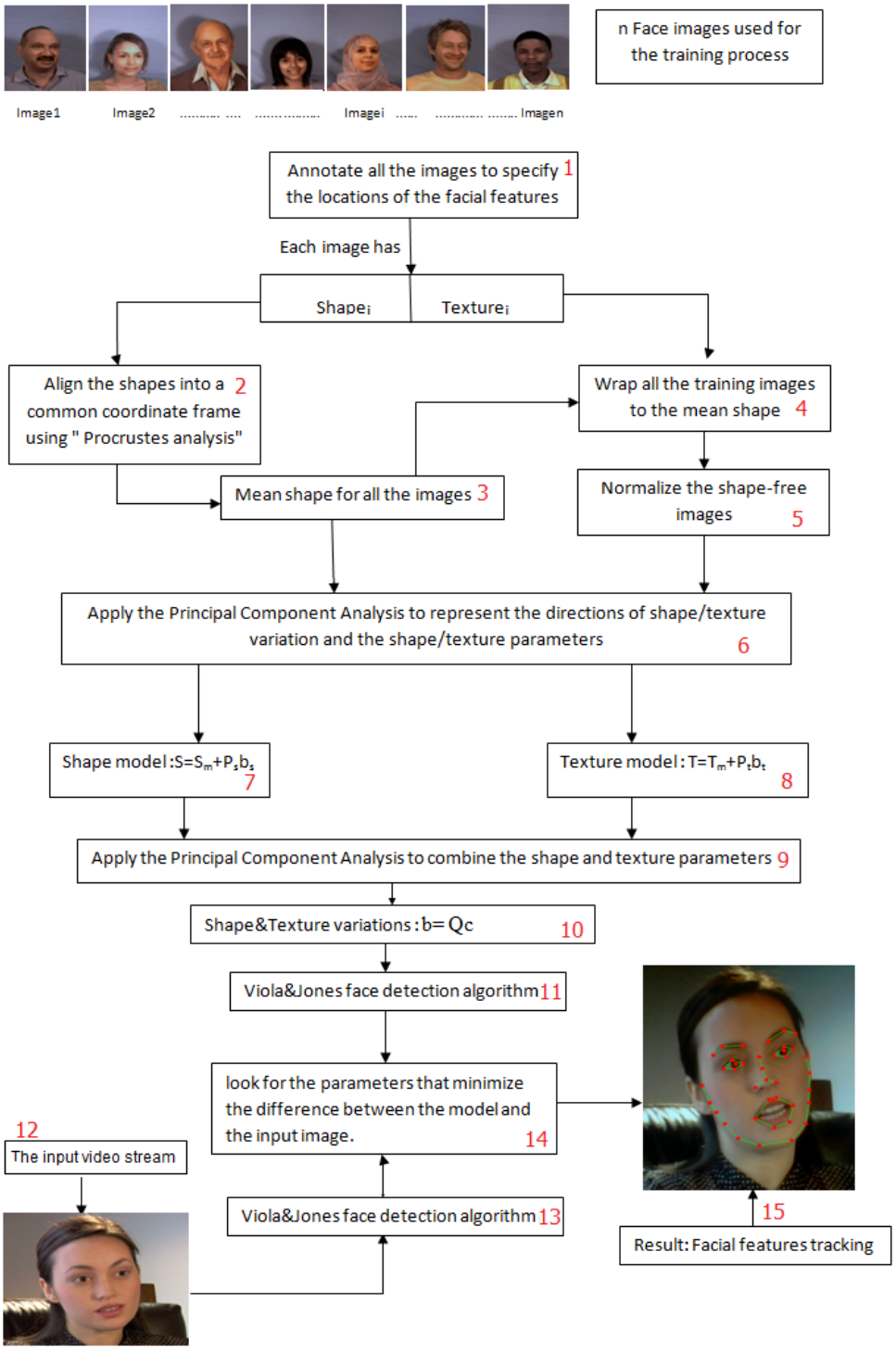


FIGURE 3.6: A pipeline of the facial features tracking solution

Figure 3.6 shows a summary of the facial features tracking method that we will develop in the next chapters. This represents our marker-less system for face facial features tracking.

Conclusion

After describing, theoretically, our chosen facial features tracking method, in the next chapter, we are going to start the application development step by, first specifying the requirements of the tracker system that we opt to implement.

Chapter 4

Requirements analysis and specification

Introduction

In this chapter, we will first describe the functional and non-functional requirements of our project. Then, will explain the dynamic behaviour of the system, by introducing the interaction between the user and the entire system.
For better clarity, the semi-formal language UML will be used to represent these requirements.

4.1 Functional requirements

The functional requirements describe the features that the system should provide to the user. In the following paragraphs, we will explain them based on the different steps of our tracking approach.

4.1.1 Annotate the dataset

The image annotation is the pre-processing step described in our tracking method. To do so, the developed application should provide the following needs:
- Annotate MUCT dataset: the application must enable the user to draw connections on the MUCT dataset.

- Annotate CDRIN dataset: the application must enable the user to draw landmark points and connections on the images captured for the CDRIN training dataset.

4.1.2 Observing the Tracking of facial features

In this functionality, the system should allow the user to see the result of the facial features tracking step.
That will be done using the training information stored in the tracker file. Therefore, in order to easily achieve the desired results of tracking, two options must exist:
- The user can choose to use the tracker file of an anterior training process.
- The user can choose to start a new training step, then use its produced tracker file.
The tracker file contains the information of the training process which can be done by building the shape and texture models from the annotated database of image, then detect the face position.
At the end, all the parameters of this training information must be saved in the tracker file.
After that, the application must recognize the facial features through a frame by frame analysis of the input video stream. Then, affix the virtual markers (animatable face model) on the face of the performer.

4.2 Non-functional requirements

Although the non-functional requirements do not express the functionalities offered by the system, they have a direct impact on the application performance.

4.2.1 Special action scene

The analysis of human motion is done using a set of images taken from a video stream captured with one or multiple cameras. The action area, in our case, is defined by a scene containing only one high definition camera and enough LED light sources to provide a high resolution images, since that improves the tracking quality.
Although we are trying to develop a facial tracking solution using one camera, we provided the possibility to add a multiple cameras tracking system.

4.2.2 Robustness

Robustness is an important parameter, it represents is the ability of the system to cope with errors during execution. It is then crucial to ensure a safe handle of the errors when loading data or processing. In fact, the algorithms proposed by our application consist of several steps; when loading files, testing the accuracy of the data is made. Also, error messages can be shown, if needed, to prevent the application from closing abnormally.

4.2.3 Extensibility

In order to take growth into consideration, it is possible to add new functionalities or extend the capabilities of the application without having to modify and adapt a lot of code. This is achievable thanks to the design method that we chose for the proposed software.

4.2.4 Ergonomic

The graphic interface must be ergonomically designed so that even novice user can use the software properly. We particularly paid attention to details, such as the ease of access to different functionalities, the visibility of images captured and annotated, the efficiency and simplicity to navigate through different windows.

4.3 Requirements specification

To have a better understanding of the required needs, we will model them in a formal way using the use case and a sequence diagrams.

4.3.1 Use case diagram

The use case diagram combines one or multiple actors, use cases and their relationships, in order to model the system of an application.

In the application that we attempt to implement, we have just one actor which is the user. To be more specific, the user here is a motion capture technician.

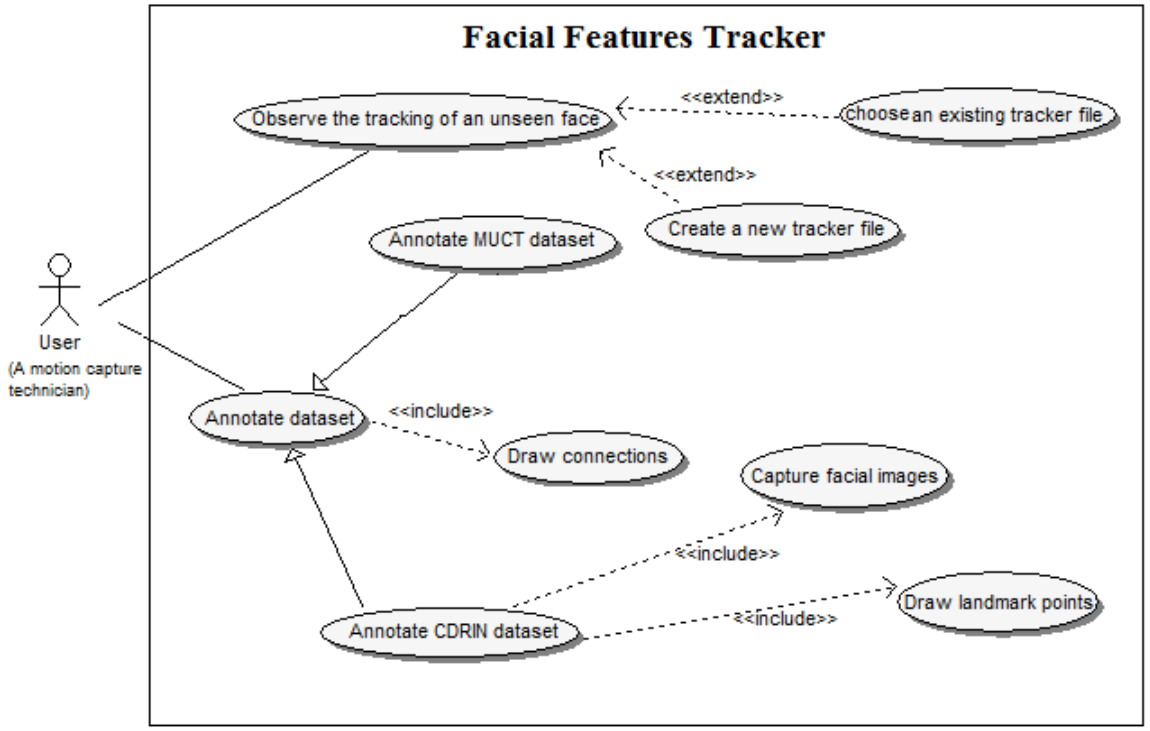


FIGURE 4.1: Use case diagram of the facial features tracking system.

The use case diagram (figure 4.1) represents the general functionalities that the facial features tracker system offers.
In order to clarify the previous use case diagram, table 4.1 presents a brief description of each use case:

TABLE 4.1: USE CASES DESCRIPTION

Use case	**Description**
Observe the tracking of an unseen face	The application should enable the user to observe the tracking of the facial features of persons from a video sequence.Unseen faces are new faces in regard to the images of the training dataset.

Table 4.1 – *Continued from previous page*

Use case	**Description**
Create a new tracker file	The "tracker" file is a file containing all the training information of the face images dataset. In this case, the application should first, build a model describing the shape variations of the training dataset. Then, build a model that describes the texture variation of the faces in the training dataset. This file will also save the information of the position of the faces in the training dataset. Creating a new file, means starting a new training process with an annotated dataset.
Load an existing "tracker" file	The user is invited to load from his computer, an existing "tracker" file that contains the information of a previous training process. After loading this file, the system will directly start the facial features tracking process.
Annotate MUCT dataset	The application should provide a tool to annotate the MUCT dataset. It is important to note that both of this use case and the next one, offer almost the same functionality, therefore, we modeled an inheritance relationship between each one of them and the annotate dataset use case.
Annotate CDRIN dataset	The application should provide a tool to annotate the CDRIN dataset that contains face images of the video games actors.
Capture facial image	The application should capture facial images from a high definition camera, in the interest of building the CDRIN dataset.

Table 4.1 – *Continued from previous page*

Use case	Description
Draw landmark points	The application should draw landmark points on the dataset images, in order to specify the features to track.
Draw connection	The application should draw curves between the landmark points to visualize the tracking results in a better way.

4.3.2 Sequence diagrams

To illustrate a part of the use case diagram defined above, we represent, in this section, the sequence diagrams of some use cases. These diagrams will explain the interaction between the user and the entire system.

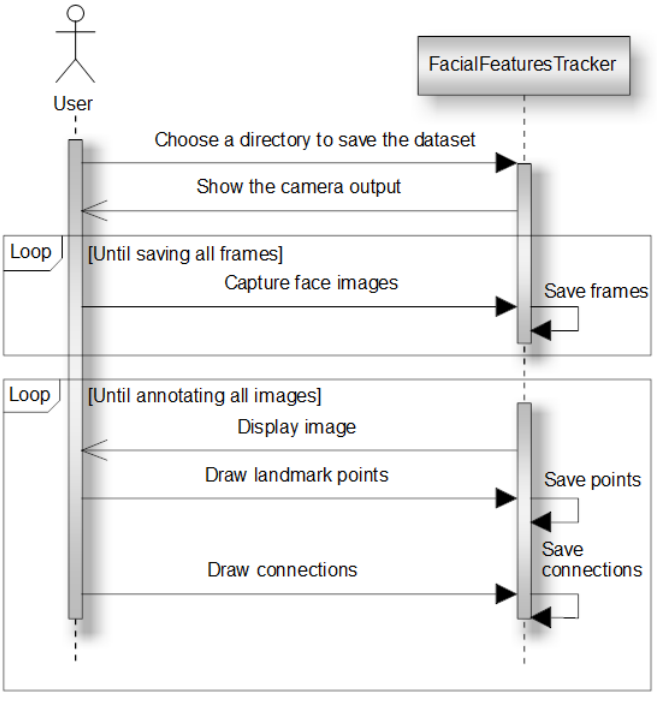


FIGURE 4.2: Sequence diagram of the person-specific dataset (CDRIN dataset) annotation use case.

Figure 4.2 shows the person-specific dataset (CDRIN dataset) annotation process. When the user chooses to create a person-specific database, he will be asked to select

the directory where all the training data will be stored in it. Then, the user will be able to capture the face images that he wants to use as a training dataset. All the captured images are simultaneously saved in the chosen directory.
Once all the needed images are captured, the annotation process starts, and the user will be invited to pick points and connections on the facial images. Then, all the coordinates of these points and connections will be saved in a specific file.

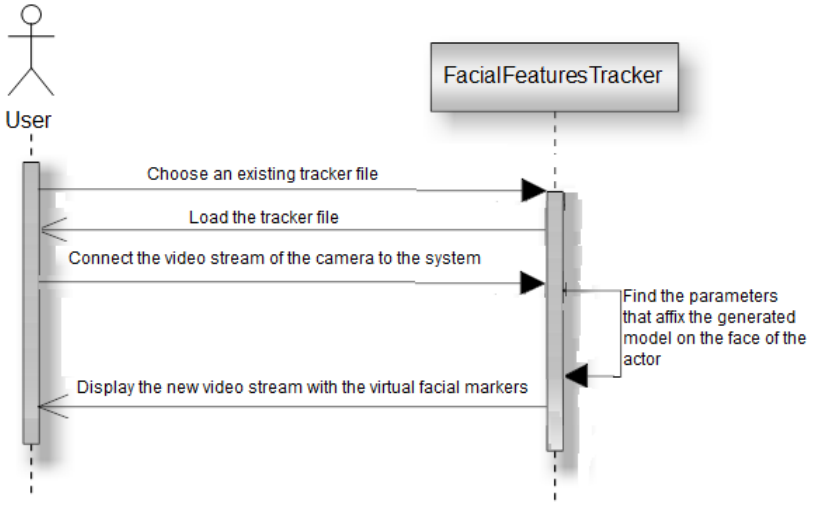


FIGURE 4.3: Sequence diagram of the use case of the tracking process using an existing tracker file.

Figure 4.3 shows the interaction of the user and the system while tracking the video games actorș facial features using an existing tracker file.
The tracker file chosen by the user here, is a file that contains the information of a previous training process. That means the information of the face detection, as well as, the shape and texture models stored from a previous mocap session.
Then, the user(a motion capture technician) connects the camera to the computer. After that, the performance of the video games actor filmed by the camera, will be displayed simultaneously on the screen but with the virtual markers affixed on the actorș face. This can be done after computing the combined model (shape and texture) parameters that best fit the new input video stream of the performer with the model. The face detection information that indicates the initial position of the face in an image, will be very helpful, because it eases this step of calculation.
Virtual markers are the set of landmark points that represent the generated models of shape and texture. In fact, each point of these virtual markers has the mission to track a specific facial feature.

Conclusion

Throughout this chapter, we have clearly identified the system actor and the functionalities that the software should offer. Therefore, the next chapter will describe the application design.

Chapter 5

Design and structure

Introduction

After finishing the use cases analysis, we focus in this chapter on designing a suitable structure for the application. This step is crucial in the course of the project and aims to undertake and prepare the ground for the implementation step.

5.1 The object oriented design

The tracking approach that we detailed in chapter 3 consists of two components: training data and algorithms. The algorithms operate on the input data (unseen faces) by relying on the stored images (training dataset) as a reference. As such, it is recommended to choose the object-oriented approach that couples the data with the algorithms through the principle of encapsulation.
Moreover, the object oriented approach provides a technique to solve the reusability problems by fundamental mechanisms such as inheritance and polymorphism. It, also, facilitates the maintenance of the application and helps to create a powerful software that is versatile, customizable and expandable easily.
After the choice of an object-oriented design, it is important to model the designed information. To do so, we opt for the UML which is a standardized and expressive way to visualize a system's architecture.
The following sections will describe the details of our application design.

5.2 Class diagrams

We expose, in this section, the application design by modeling the class diagrams that will help us to better understand the mechanism of the implementation of the software. For clarity purposes, we mentioned the most important attributes and methods of the classes.

5.2.1 Class diagram of the annotation stage

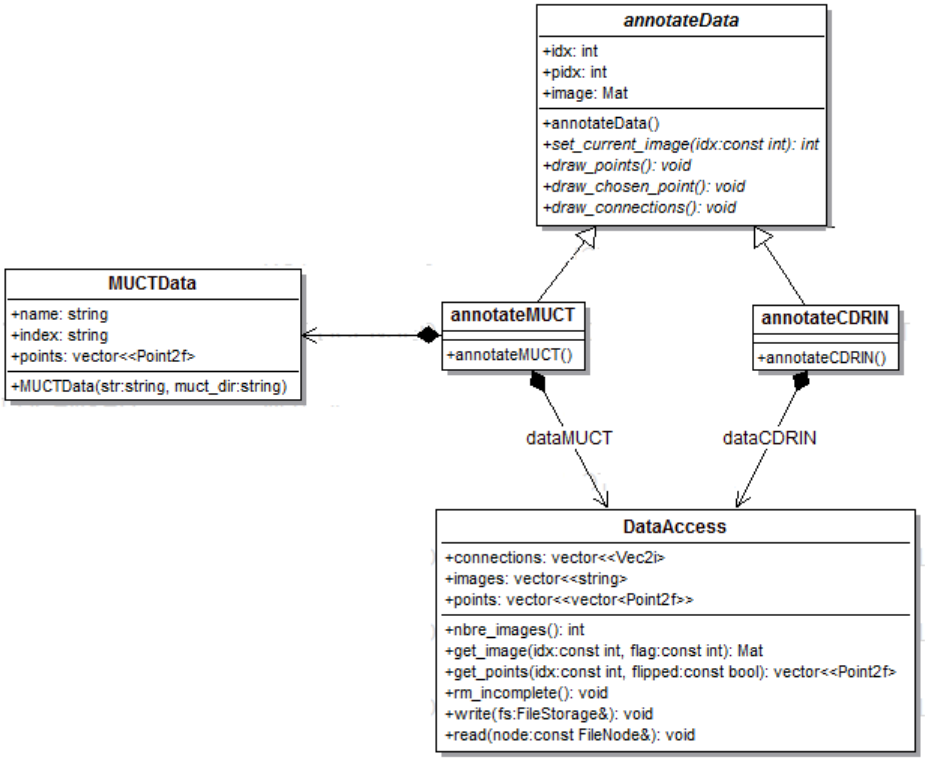


FIGURE 5.1: Class diagram of the annotation stage.

At this pre-processing stage, we are interested in annotating our two training datasets. Therefore, we declared the annotateMUCT and annotateCDRIN classes that correspond respectively to the annotation of the MUCT database (introduced in chapter 3) and the annotation of the specific face images captured in the CDRIN studio. Considering that both of these classes have almost the same role, so, they inherit from the annotateData abstract class which allows to draw the points and the curves on the dataset images.
It is important to mention that the MUCT data is a pre-annotated dataset. However, it lacks connections between the facial feature points and this is done by the annotate-MUCT class which uses the MUCTData class to access to the images and the annotated

points.

The DataAccess class is responsible for managing the dataset, and implements the storage and access functionality.

5.2.2 Class diagram of the training and tracking stage

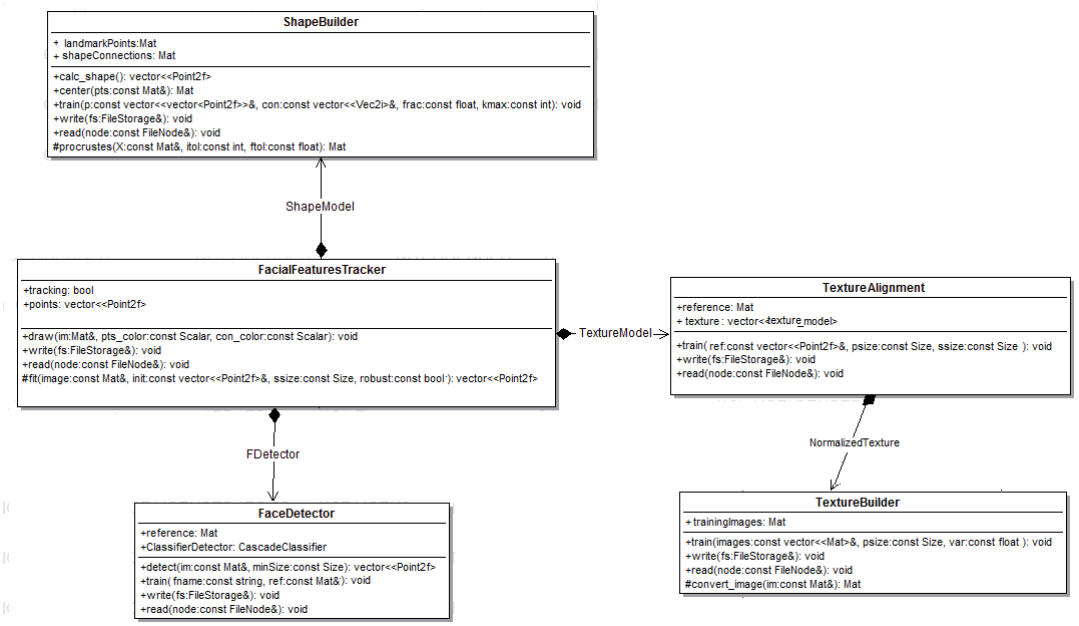


FIGURE 5.2: Class diagram of the training and tracking stages.

After annotating the dataset, we are looking to start the processing stage. To do so, we created the FacialFeaturesTracker class, described in figure 5.2 that tracks the features of an unseen face, using the FaceDetector, the ShapeBuilder and the TextureAlignment classes corresponding respectively to the class describing the model initialization method (explained in chapter 3), the class responsible of building the shape model of the face and the class that creates a texture model from the constructed Normalized Texture of the TextureBuilder class.

As we can notice from this diagram, the relationship between the classes is, an aggregation which is a special type of association in which objects are assembled or configured together to create a more complex object.

5.3 Graphical user interface structure

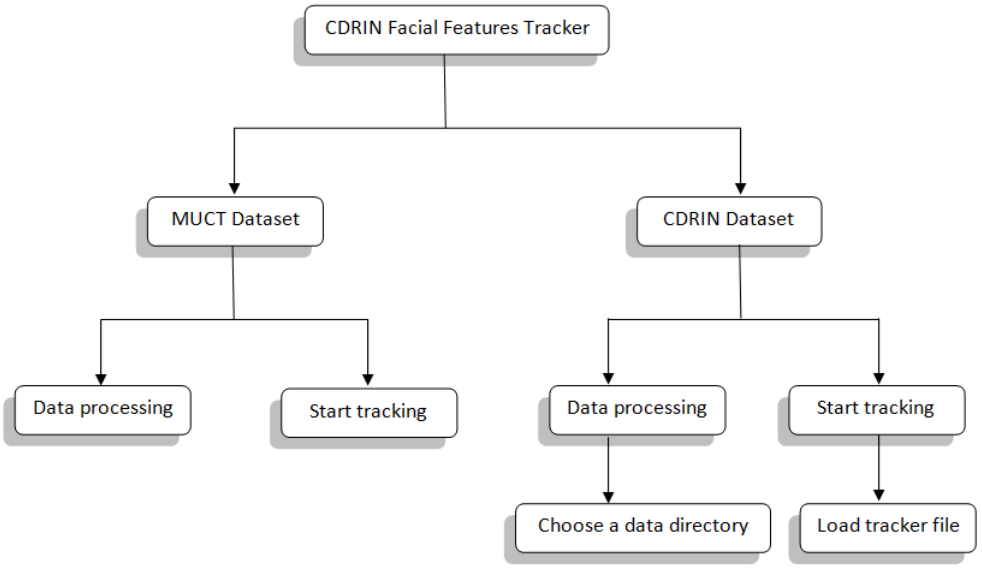


FIGURE 5.3: The GUI structure of the facial features tracker application.

In order to provide a better human-computer interaction while using the application, we decided to develop a graphical interface that guides the user and allows an easy tracking process.
The diagram discribed in figure 5.3 shows the structure of the different windows of the application GUI, and we will expose, in the next chapter, some screenshots of these windows.

Conclusion

In this chapter, we presented the design of our solution, in order to facilitate the transition to the implementation phase. In the next chapter, we will show the achieved work as well as the technical tools that we used for the development.

Chapter 6

Implementation and results

Introduction

This chapter focuses on the implementation of the application's components. We begin by presenting the development tools, then we present the accomplished work. At the end, we evaluate the tracking performance through experimental results, and a test of the influence of the training dataset choice.

6.1 Development environment

In this section we present briefly the technical tools that we chose, in order to develop our application.

6.1.1 Hardware tools

The algorithms depend on the computing power, the availaible memory and the central processing unit time. The materials that have been used to develop the facial features tracking solution, have the following features:
- Operating system : Windows 7.
- Processor : Intel(R) Core(TM) i7-2670QM CPU @ 2.20GHz GHz.
- RAM, Hard Disc : 6,00 GO, 640 GO.
- Capture tools: Replay XD1080 HD video action camera and LED light sources.

6.1.2 Software tools

The choice of the implementation language

As we explained in the non functional requirements (chapter 4, section 4.2.3), it is important to choose an object oriented language that proposes polymorphism and inheritance and facilitates the addition or the modification of the classes. Therefore, the C++ language is the best choice, since it provides all these options. Moreover, it is prevalent in the field of digital imaging and computer vision.

To implement the graphical user interface, we will use the Microsoft Foundation Class (MFC), to create the interface, and facilitate the data access and the result visualization. It is important to note that the development of our application was done using the visual studio 2012 IDE.

The choice of the OpenCV library

The algorithms that we attempt to implement require matrix computations and some image processing techniques. So we need to seek for a solid open source library with C++ interface.

It turned out that the OpenCV Library, which is a library proposed by Intel and optimized for Windows, has a complete C++ matrix interface, with everything needed for basic algebra operations. Also, OpenCV could take care of converting between matrices and images matrices.

Therefore, we chose the OpenCV2.4.5 version, as it includes a large number of functions that match our needs.

Moreover, this library provides some interesting books that demonstrate how to quickly build computer vision applications such as [1], and that helped us a lot in the implementation stage.

6.2 Results

In this section we illustrate the implementation result through some screenshots of the CDRIN Facial Features Tracker application.

Despite our project aims to track facial features of video games actors, for validation purposes, we will use the talking face video made available from the live action set of the digital Emily project [32], as a facial image database for CDRIN.

6.2.1 The home window

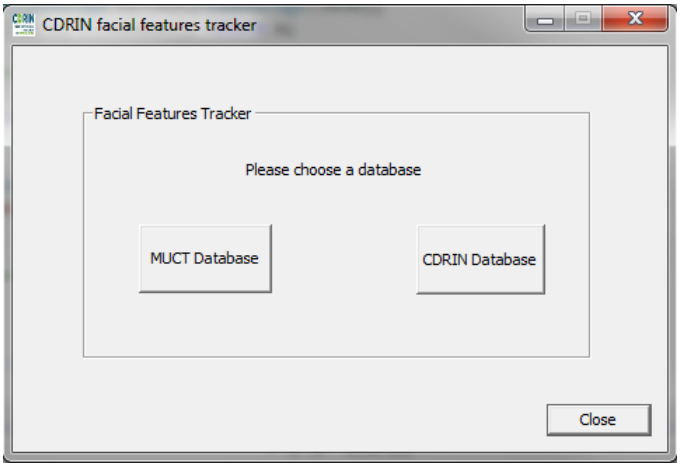


FIGURE 6.1: Home window of the CDRIN Facial Features Tracker.

This is the main window of our developed application, where the user is invited to choose a the dataset of facial images that he wants to use for the training and the tracking process.

6.2.2 Annotation tool

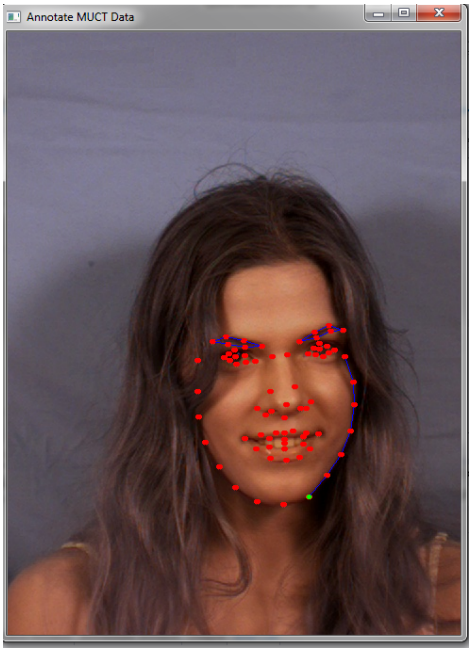


FIGURE 6.2: The MUCT dataset annotation.

In figure 6.2, we were about to draw the connections between the landmark points of the first image of the MUCT pre-annotated dataset.

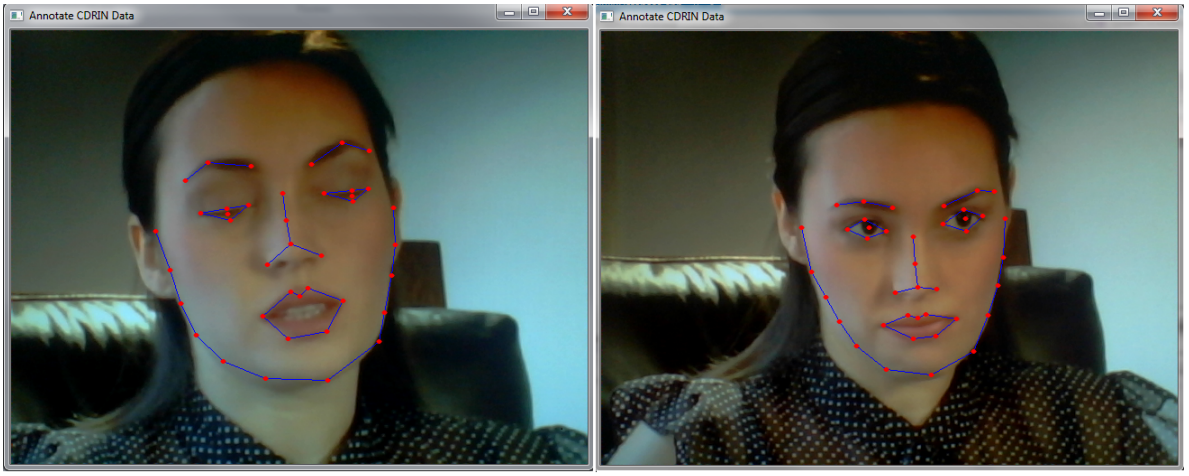


FIGURE 6.3: Two annotated training images from the 8 facial images used for this test.

Figure 6.3 illustrates the landmark points and the connection annotation on the images considered as a CDRIN dataset.

6.2.3 Training process

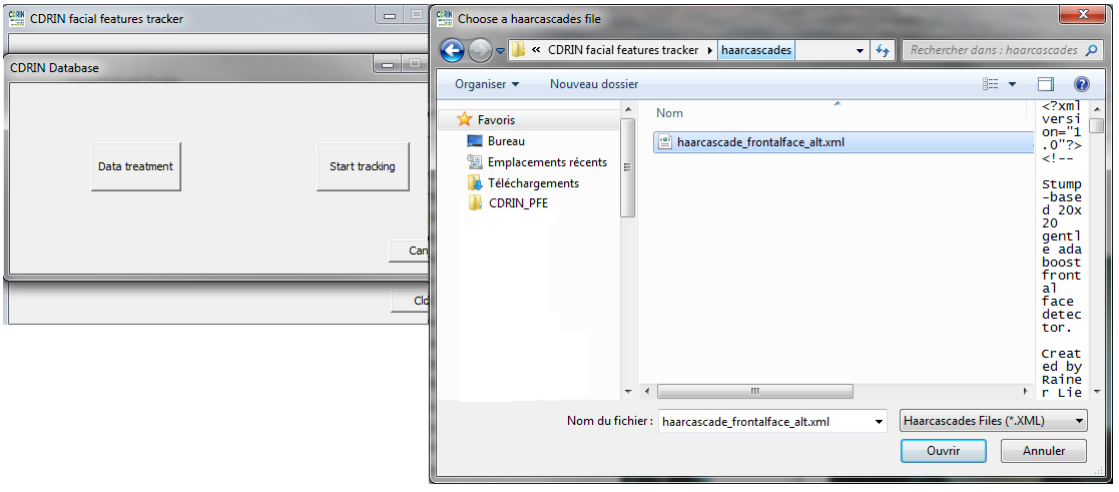


FIGURE 6.4: Selecting the haarcascades-frontalface file.

After the annotation step, the training process starts and the user will be asked to select the haarcascades-frontalface file used for the face detection algorithm (figure 6.4).



FIGURE 6.5: The "wait-window" of the training process.

Once the file is selected, the user will be invited to wait for a period of time in order to fulfill the training of all the images in the dataset (figure 6.5). In fact, this period depends on the number of images in the training dataset, the number of the annotated points and the power of the processor used for this process (please see section 6.3.1 for further details).

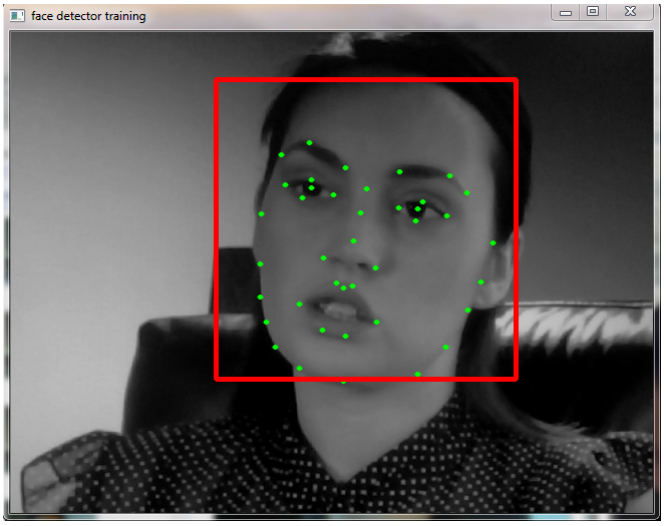


FIGURE 6.6: The training of the face position using the haarcascades-frontalface file.

The training of the face detection is one of the steps of the training process (figure 6.6). It comes after building the shape and the texture models. We notice here that the training images are converted to a single-channel images, in order to ensure robustness against differences in contrast and changes in illumination conditions.

6.2.4 Tracking process

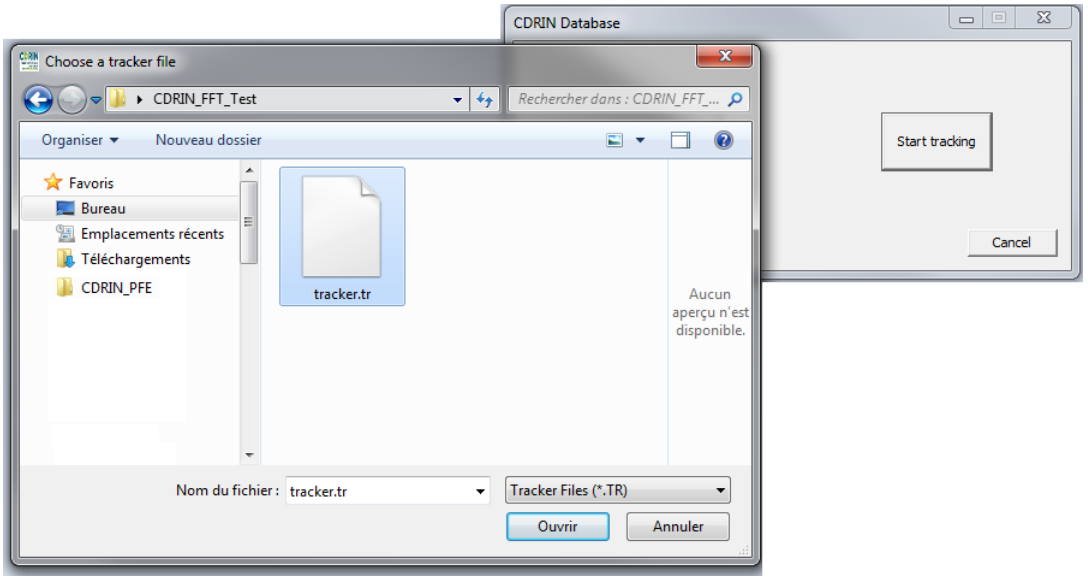


FIGURE 6.7: The tracker file selection.

At this stage, The user is asked to select the tracker file containing the training information of the dataset that he wants to use for tracking.
After that, a new window appears and displays the video stream of the face to track.
The following sections show the result of the face tracking and animation of a 3D character. We studied also the effects of using two different training datasets, on the tracking results.

6.2.5 A 3D character facial expressions animation

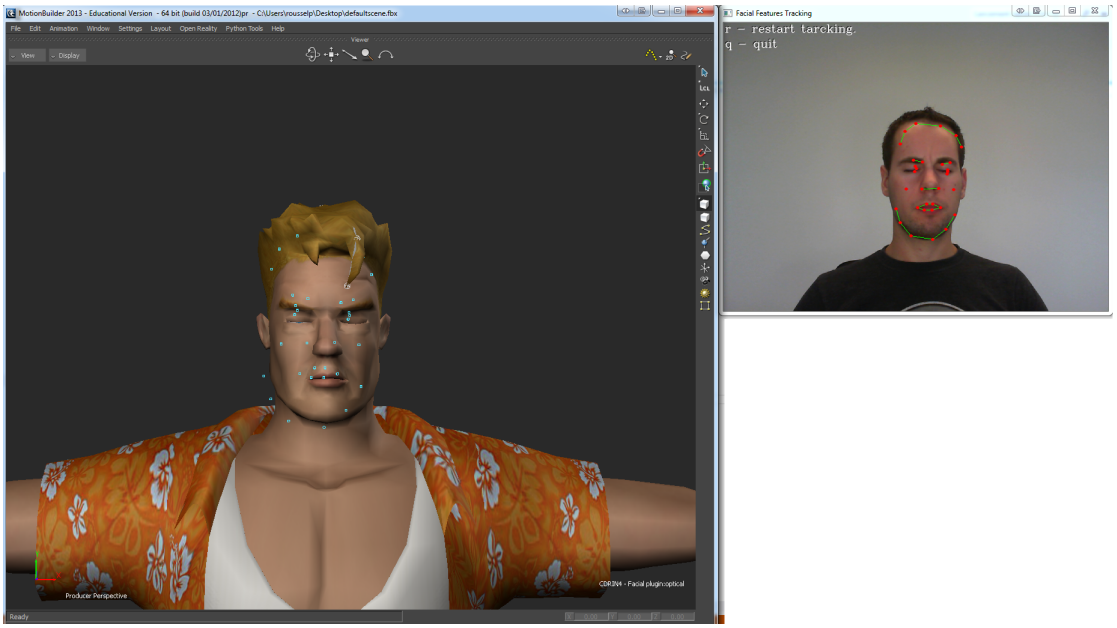


FIGURE 6.8: An example of eyes tracking (right) and animation with the MotionBuilder software (left).

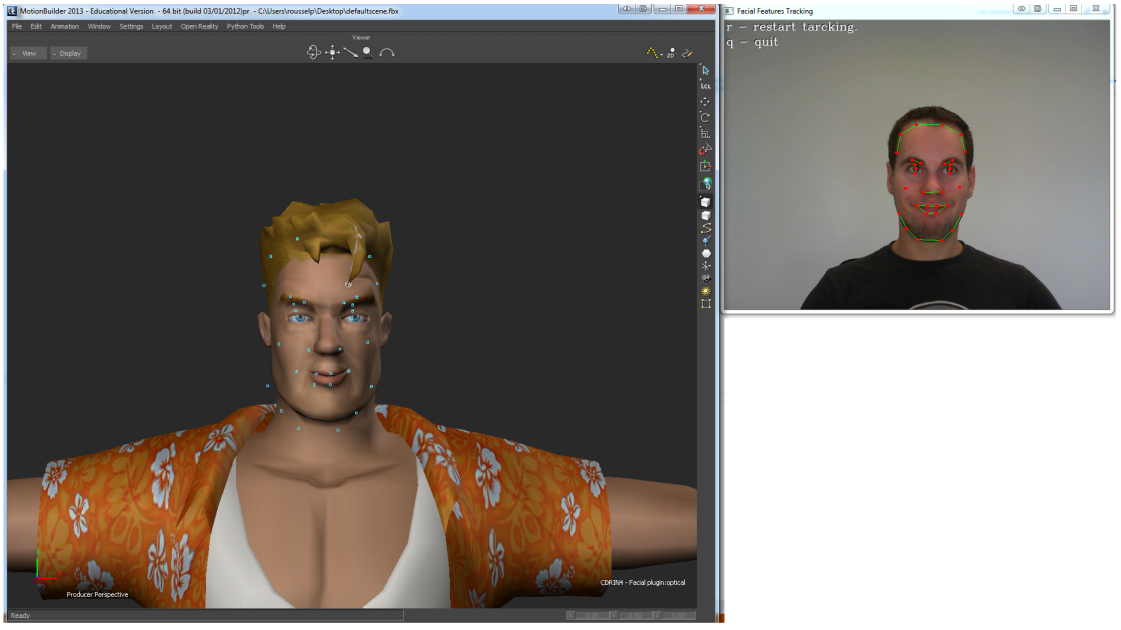


FIGURE 6.9: An example of mouth tracking (right) and animation with the MotionBuilder software (left).

This is an application of our facial features tracking solution on a the facial animation of a 3D character.

Figures 6.8 and 6.9 show the result of facial animation for a 3D character using our developed set of virtual markers. As we can see, the tracking is displayed using our facial features tracker software and the animation of the 3D character is done in real time using the MotionBuilder software. It is important to note here, that the 3D character was offered by the CDRIN team.

6.3 Performance evaluation

In the following sections, we show some experimental results and comparisons, in order to evaluate the performances of our developed facial features tracker.

6.3.1 Experimental results

In this section, we present the effect of varying the number of landmark points and the number of training images, on the tracking robustness and the training execution time[1]. The tracking robustness is measured, here, based on the frame-rate at which the tracker model affixed on the actor's face is displayed on the screen.
Since it is difficult to simultaneously understand the effect of two variables, we divided our experiments into two parts and we fixed the value of one variable for each set of tests.

[1]The processor used for all the tests is an Intel(R) Core(TM) i7-2670QM CPU @ 2.20GHz GHz.

6.3.1.1 The analysis of tracking results and training execution time using different numbers of landmark points and a fixed size of the training dataset

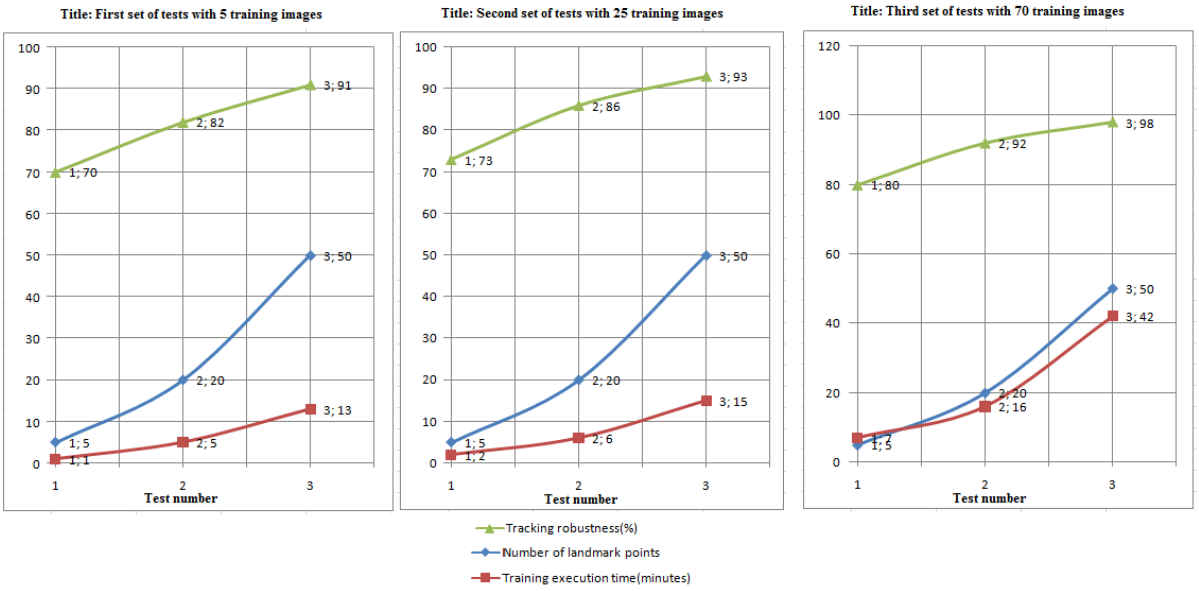


FIGURE 6.10: Three sets of experimental tests using our developed facial features tracker with a fixed number of training images.

Figure 6.10 shows the results of three experiments using respectively 5, 25 and 70 training images. We plot for each test, the training execution time (red curve), the tracking robustness (green curve) and the number of landmark points (blue curve). It is immediately apparent that the bigger the set of landmark points is, the more robust the facial features tracking becomes. However, increasing the number of landmarrk points carries with it a significant increase in the training execution time. This can be explained by the important computational complexity needed to align and wrap the landmark points of the training dataset.

6.3.1.2 The analysis of tracking results and training execution time using different numbers of training images and a fixed number of landmark points

Figure 6.11 shows the results of three experiments using respectively 5, 25 and 70 landmark points. We plot for each test, the training execution time (red curve), the tracking robustness (green curve) and the size of the training dataset (blue curve).

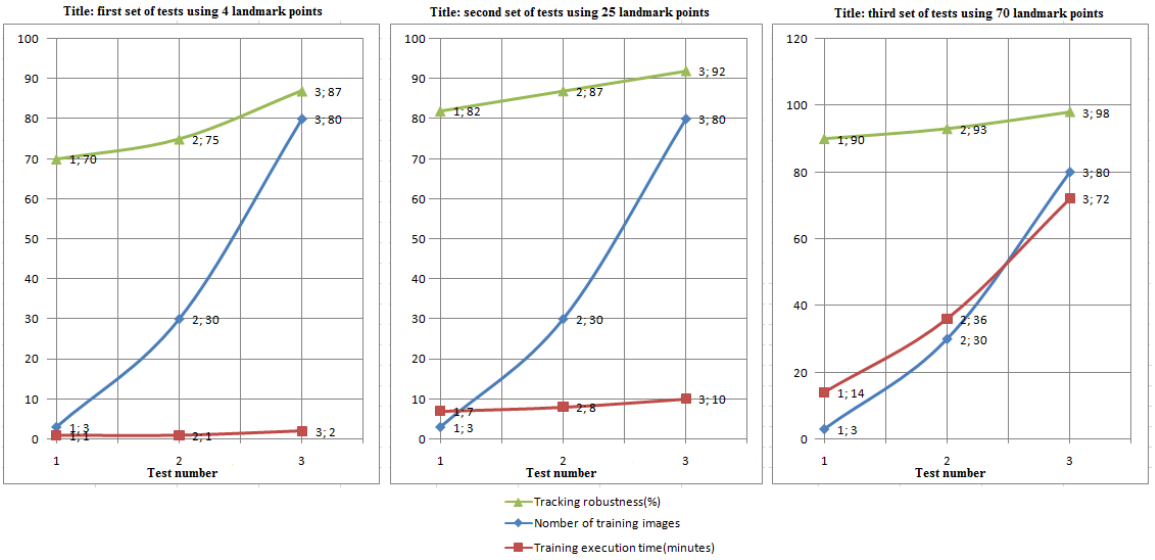


FIGURE 6.11: Three sets of experimental tests using our developed facial features tracker with a fixed number of landmark points.

Such as in figure 6.10, we can notice that the bigger the training dataset size is, the more robust the facial features tracking becomes. But with these tests, we find also the same problem of tracking robustness versus the long training execution time.

6.3.2 The tracking results of generic and person specific training

In this section, we are trying to test the influence of the training dataset choice on the performance of the implemented facial features tracking method.

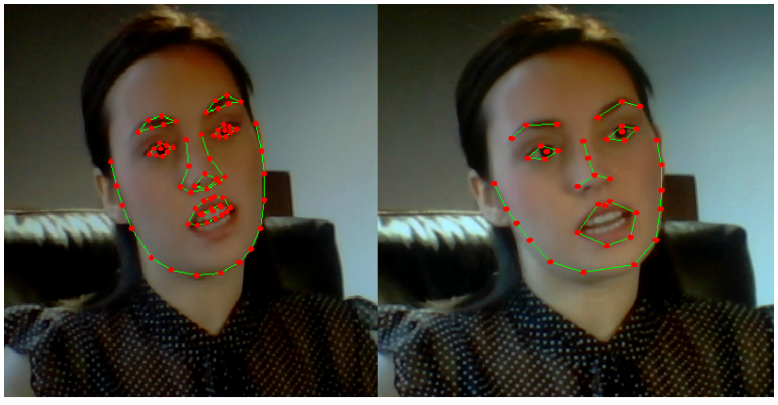


FIGURE 6.12: The result of a generic (left) versus person specific (right) training.

In figure 6.12, the tracker model of the left image was trained using the MUCT dataset. However, the tracker model illustrated in the right image was learned from the set of images captured for the CDRIN specific database.
As we can see, the facial features tracking using a person specific training dataset is more efficient. In fact, the purpose of the person specific tracking is to describe the facial feature movements of one particular person, the one it has been trained for. Whereas, the generic training is able to model faces of different persons but the tracking using this type of training, appears to struggle even for simple facial expressions.

Conclusion

In this chapter, we presented the development environment in which we created our application, the used language and techniques. We, subsequently, showed the most significant windows of our application, as well as its performance evaluation.

Conclusion and future prospects

During this graduation internship, we worked with the Digital Imagery Research and Development Centre team, and we learned a lot from this great experience, which made me ready to pursue new professional challenges.
The aim of this project was to improve the facial tracking system of the CDRIN by developping a markerless solution for the facial features tracking of video games performers. This new tracking system facilitates the facial motion capture of the actors which leads to produce a better animation results of the video games characters.

To do so, we began with an exhaustive research in the fields of motion capture, digital imaging and computer vision in order to find the method that is most suitable for facial features tracking. There is much literature about face detection and features extraction, and studying all their methods is not a trivial task. Therefore, we presented, in the second chapter of this report, the most popular approaches related to these topics. Then, we performed a comparison of the classical facial features extraction techniques. All this research work, hepled us to find an efficient facial features tracking solution, basically inspired from the statistical model approach. The facial features tracking method explained in chapter 3, consists of four main steps:
- Annotate a database of facial images using curves and landmark points, in order to specify the locations of the facial features to track for each mocap session.
- Build the deformable model that is able to describe the shape and the texture parameters of the training dataset.
- Detect the face position in the image to facilitate the tracking process.
- When the video games actor starts his performance, the shape and texture data stored from the previous steps will be used in order to find the parameters that best affix the deformable face model, previously generated, on the real face of the actor, filmed with the camera.

In order to visualize and evaluate the results of this facial features tracker, it was necessary to develop a software using the C++ language and the OpenCV library. The

technical details about the design and the implementation were described in the last three chapters of this report.
The final results of face tracking and 3D character animation were presented, in the last chapter, with a detailed study of the performance evaluation of our developed facial features tracker.

It should be noted that the method for face tracking used in this project can work effectively in constrained settings. However, we can improve the tracking results by using a calibrated multi-cameras to capture the facial expressions of the actor from multiple angles. Moreover, it is possible to speed-up the process of model training and reduce its computational cost by using multiple PCs to parallelize the process, or by using the graphics processing unit computing power.

Appendix A

Performing PCA by the Singular Value Decomposition method

First of all, we assume that the data has been preprocessed to have a zero mean.
We decompose the X matrix using SVD, i.e.

$$\mathrm{X} = \mathrm{US}^t V$$

Since V is an orthogonal matrix, then $\mathrm{V}^t V = I$ and we can write the covariance matrix as follows :

$$\mathrm{C} = \frac{1}{n} X^t X = \frac{1}{n} U S^{2t} V$$

In this case U is a n x m matrix. Following from the fact that SVD routine order the singular values in descending order we know that, if n ¡ m, the first n columns in U corresponds to the sorted eigenvalues of C and if m¿= n, the first m corresponds to the sorted non-zero eigenvalues of C.
The transformed data can thus be written as

$$\mathrm{Y} = \mathrm{P}^t X =^t PPS^t V$$

where ${}^t PP$ is a simple n x m matrix which is one on the diagonal and zero everywhere else. To conclude, we can write the transformed data in terms of the SVD decomposition of X [43].
We notice that SVD is closely related to PCA in terms of computing eigenvalues and eigenvectors. From the SVD calculation, the PCA output is calculated by the outer product of the columns of U with the singular values diag(S) [17].

Appendix B

Wraping texture to the mean shape

In order to warp a texture, so that a set of control points are mapped to new positions in the mean shape, the texture is mapped in a way that the control points from each sample match the control points of the mean shape.
This concept is very simple, we will create triangles including our annotated points and then map from one triangle to another.
There is a well known method in mathematics and computational geometry to do this. It is the delaunay triangulation, where no point of the set of landmarks is inside the *circumcircle* (the circle that has all the three triangle vertexes) of any triangle in the triangulation.

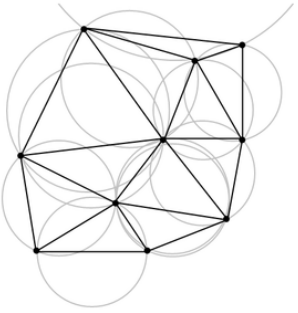


FIGURE B.1: A delaunay triangulation with visible circumcircles. (Source: [N10])

After mapping each pixel inside a triangle of a face image into its correspondent triangle in the mean shape, the mean shape control points will be the reference points since all the texture is processed on this reference shape.

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